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GIST: Groundwater Interpolation in Space and Time

Spatiotemporal Groundwater Level Forecast Interpolation
for Brandenburg Using Gated Recurrent Unit and Gaussian Process

Master Thesis

submitted by

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Abstract

Groundwater monitoring networks cover only a fraction of the locations where predictions are needed, and most forecasting systems work only where wells already exist. Forecasting groundwater levels (GWL) at existing monitoring wells improves considerably with deep learning. However, predicting GWL at entirely unmonitored locations without any historical observations is more difficult and receives less attention. Most existing approaches do not test against large, fixed test sets of spatially separated wells and therefore cannot "forecast anywhere".

This thesis introduces GIST (**G**roundwater **I**nterpolation in **S**pace and **T**ime), a spatiotemporal pipeline combining a temporal and a spatial modeling component, covering 1,040 monitoring wells across Brandenburg, Germany. A Gated Recurrent Unit (GRU) produces temporal forecasts at monitored wells, which a Gaussian Process (GP) then interpolates to locations excluded from training to simulate unmonitored locations.

The GRU matches the performance of a much larger Temporal Fusion Transformer baseline while training roughly 90 times faster, which makes the full experimental setup feasible. One of the main findings is that temporal forecast quality is not the bottleneck in the spatiotemporal pipeline: interpolating the actual observations at training wells instead of the temporal predictions does not improve the error of the full spatiotemporal pipeline. The limiting factor is therefore spatial generalization, not temporal forecast quality. Distance to the nearest training well is the strongest predictor of per-well error. A jointly trained model combining GRU and GP with a combined loss does not improve overall performance but specifically reduces error for the most spatially isolated wells, which are the hardest cases for a "forecast anywhere" system. Together, these results demonstrate the feasibility of spatiotemporal GWL forecasting at unmonitored locations and show that further progress mainly depends on improving spatial generalization.

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1. Introduction

Groundwater supplies water for drinking, agriculture, and ecosystems worldwide (Taylor et al., 2013), but monitoring networks cover only a fraction of the areas that depend on it. The German state of Brandenburg is characterized by a continental climate, low annual precipitation (500–700 mm (Deutscher Wetterdienst, 2024)), declining groundwater recharge (Francke and Heistermann, 2025), and extensive agricultural (49%) and forested (35%) land (Amt für Statistik Berlin-Brandenburg, 2023). Since both agriculture and forests rely heavily on groundwater, and the low precipitation limits natural recharge, groundwater availability is crucial for agricultural productivity and forest health. Reliable groundwater level (GWL) forecasts are therefore essential for farmers, authorities, land managers, and planners.

The Brandenburg State Office for the Environment (LfU) operates a dense network of over 2,100 monitoring wells (Landesamt für Umwelt Brandenburg, 2024), of which 1,040 remain after quality filtering. The wells are recording weekly GWL observations that span more than seven decades. These wells are spread across Brandenburg’s full territory but naturally cannot provide continuous spatial coverage, which leaves the vast majority of areas effectively unmonitored. The key challenge this thesis deals with is:

Given past groundwater level observations at monitored well locations, can we reliably predict groundwater levels at unmonitored locations at future horizons?

The long-term goal is reliable forecasting at unmonitored locations, in other words a “forecast anywhere” system.

For Brandenburg’s farmers, foresters, and water managers, knowing what the groundwater level will do over the next few weeks at their specific location naturally matters more than knowing what it does at the nearest monitoring well. Decisions about irrigation scheduling, emergency groundwater pumping during drought, and long-term land management all depend on local conditions. The LfU network is dense by any standard, but roughly 2,100 wells (of which 1,040 meet the quality criteria for this study) still cannot cover 29,654 km² continuously. A practical “forecast-anywhere” deployment requires reliable GWL estimation at any point in the state on demand.

This problem has two dimensions:

Temporal Predicting future GWL at a well given its history and input features

Spatial Interpolating temporal GWL forecasts to a location where no observations exist

Handling both simultaneously, and therefore predicting GWL at locations where no observations exist, is the core challenge of spatiotemporal GWL forecast interpolation as formulated in this thesis.

1.1. Problem Statement

Formally, the training data consist of time series $y_{i,t}$ of groundwater levels at wells $i \in \mathcal{W}_{\text{train}}$ for time steps $t = 1, \dots, T$. Given a set of dynamic and static features, our goal is to predict the future GWL time series $\hat{y}_{j,t+1:t+H}$ at test wells $j \in \mathcal{W}_{\text{test}}$, where $\mathcal{W}_{\text{train}} \cap \mathcal{W}_{\text{test}} = \emptyset$. Forecasts are produced at all forecast horizons from 1 to $H = 16$ weeks ahead.

At the spatial training wells $\mathcal{W}_{\text{train}}$, the time series are further divided into a temporal training period, a validation period, and a temporal test period. The temporal test period also provides the true observations used by the true-observation interpolation baseline.

The key challenge that distinguishes this problem from standard time series forecasting is that the test wells are entirely unseen by the model during training. Spatial interpolation comes entirely from learned relationships between hydrogeological features and GWL dynamics.

1.2. Contributions

Reliable spatiotemporal GWL estimation at unmonitored locations would matter for water management and drought response, but whether such estimation can be done reliably is not obvious. The spatial interpolation step could fail because temporal forecasts are too noisy, because the GP cannot generalize across the spatial structure of GWL in Brandenburg, or because no available hydrogeological feature carries enough information to bridge the gap between monitored and unmonitored locations. This thesis investigates these potential failure modes directly. That the GRU can match the TFT at a fraction of the training cost is not a contribution in isolation; it is what makes everything below possible. The contributions are:

1. A **GRU sequence-to-sequence model** that matches the predictive performance of the Temporal Fusion Transformer of Kunz et al. (2025) on the Brandenburg dataset while being nearly two orders of magnitude faster to train, which enables full hyperparameter optimization (HPO), feature ablation studies, multi-seed stability analysis, joint training runs, and a large number of preliminary experiments.
2. An analysis of how **interpolation difficulty scales with spatial isolation** across 40 random split seeds, which shows that seeds drawing spatially isolated test wells are systematically harder to predict. This finding motivates the 40-seed evaluation design.

3. A **spatiotemporal two-stage pipeline** (GRU for temporal forecasting, GP for spatial interpolation), evaluated under the co-location constraint on a **random 90/10 split** of 104 test wells.
4. A **GP feature ablation study** over 30 spatial feature configurations that shows hydrogeological zone to be the single most informative covariate for spatial interpolation, with no other feature or combination matching its performance on the main 90/10 random spatial split.
5. An analysis of the **true-observation interpolation baseline** which replaces GRU predictions with true observations as GP input and shows that the gap between using true observations and GRU predictions as GP input is negligible, which means that temporal forecast quality is not the primary bottleneck for the spatiotemporal forecast interpolation problem.
6. An analysis of **distance-dependent effects in jointly trained spatiotemporal modeling**: on average, end-to-end joint training of the temporal and spatial components does not outperform the two-stage pipeline, but for test wells more than roughly 4.5 km from any training well the ranking reverses, which is the setting where the two-stage pipeline is least reliable.

1.3. Thesis Structure

Chapter 2 surveys prior work on deep learning for hydrology, GWL forecasting with machine learning, and spatial interpolation approaches. Chapter 3 describes the Brandenburg monitoring network, the static and dynamic features used as model inputs, and the temporal and spatial structure of the data. Chapter 4 introduces the GRU and GP components, the spatiotemporal pipeline designs, and the evaluation protocol. Chapter 5 reports the empirical findings. Chapter 6 interprets the results, documents approaches that did not work, and discusses the limitations. Chapter 7 summarises and points to future work.

2. Related Work

This chapter surveys the literature this thesis builds on, covering deep learning for hydrological forecasting (Section 2.1), groundwater level prediction with machine learning (Section 2.2), prediction at spatially unmonitored locations (Section 2.3), spatial interpolation and Gaussian Process regression for groundwater (Section 2.4), joint spatiotemporal modeling (Section 2.5), uncertainty quantification (Section 2.6), and reproducibility practices in deep learning for hydrology (Section 2.7). The chapter closes with a dedicated discussion of the most directly related work, Kunz et al. (2025), and the gaps this thesis fills (Section 2.8).

2.1. Deep Learning for Hydrological Forecasting

Hydrological time series modeling has been dominated by recurrent neural networks (RNNs). Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) solved the vanishing gradient problem of early RNNs by introducing memory cells and multiplicative gating mechanisms, which makes it possible for gradients to flow over long sequences. This matters for hydrology, where groundwater and streamflow dynamics show long-term patterns over weeks to months. The Gated Recurrent Unit (GRU) (Cho et al., 2014) simplifies the LSTM design to two gates at lower computational cost, which makes it a practical alternative for the long look-back windows (52 weeks) required in this thesis. Each GRU cell has its own reset and update gate, so cells can specialize across timescales: cells that mostly keep their update gate active hold onto long-range patterns, while cells that frequently reset respond to recent inputs. This makes the GRU a good fit for groundwater dynamics, where precipitation responses are fast and seasonal recharge is slow. Willard and Varadharajan (2025) systematically evaluate GRUs for prediction at unmonitored locations across a large set of stream temperature prediction tasks and find similar performance to LSTMs.

Kratzert, Klotz, Brenner, et al. (2018) show that LSTMs trained on 241 catchments from the CAMELS (Catchment Attributes and Meteorology for Large-sample Studies) benchmark dataset can match or exceed calibrated conceptual models (physics-based watershed models whose parameters are fitted to observed streamflow at each individual site) for rainfall-runoff prediction. A key finding is that models trained jointly on groups of catchments can match separate models fitted to each catchment individually. The paper further shows that LSTMs substantially outperform traditional RNNs in snow-influenced catchments: the RNN cannot retain multi-month snow accumulation across winter because of vanishing gradients, whereas the

LSTM cell state stores this information without any explicit snow encoding. This thesis uses a GRU for the same reason: seasonal groundwater recharge also develops over months.

Kratzert, Klotz, Herrnegger, et al. (2019) extend this approach to the prediction-in-ungauged-basins (PUB) problem, which asks how to predict at basins with no observed streamflow, by feeding fixed catchment features into a global model, which lets it generalize to unseen sites without per-site training. They show that an LSTM trained on gauged basins can generalize to entirely withheld basins via static catchment attributes alone, outperforming calibrated physics-based models despite never training on those basins. The PUB problem is similar to the spatial interpolation challenge of this thesis: just as PUB requires predicting streamflow at a site with no historical observations, this thesis requires predicting groundwater levels at wells where no time series has ever been recorded.

Gauch et al. (2021) extend the global model idea to multi-timescale prediction, introducing a modified LSTM architecture that simultaneously generates predictions at daily and hourly temporal resolutions.

Kratzert, Gauch, et al. (2024) argue in a HESS Opinions piece that training on one basin at a time is statistically inferior: models then overfit to local patterns and miss the signal shared across sites. This thesis follows the same reasoning.

Nearing et al. (2021) argue that the success of DL for predicting at ungauged basins is a call to action for the hydrology community to develop quantitative, scale-relevant theories of watersheds alongside ML.

2.2. Groundwater Level Prediction with Machine Learning

Compared to surface water modeling, groundwater level prediction differs in a few ways: slower seasonal dynamics, strong coupling to geology and land use, and sparser observational networks. Tao et al. (2022) review over a hundred studies applying feed-forward networks, SVMs, random forests, LSTMs, and hybrid models to GWL prediction across diverse hydrogeological settings. Most published work only evaluates on test time periods at monitored wells, so the harder problem of predicting at unmonitored locations is rarely addressed.

Wunsch et al. (2021) systematically compare several neural architectures for GWL forecasting across a set of monitoring wells in the Upper Rhine Graben, spanning Germany and France. NARX models outperform LSTM and CNN architectures on most metrics in both single-step and multi-step forecast settings, though for multi-step forecasting LSTMs and CNNs achieve higher R^2 scores.

Heudorfer et al. (2024) train a global LSTM conditioned on site-specific attributes on 108 monitoring wells distributed across Germany, combining dynamic meteorological inputs with static site attributes, and achieve a mean NSE above 0.8 in an in-sample setting, which is

comparable to or better than single-well models trained separately on each location. Their study is the direct predecessor to Kunz et al. (2025), which scales the approach to 5,288 wells using TFT and N-HITS architectures. The more informative result comes from their out-of-sample evaluation, where entire wells are withheld from training: mean NSE drops to approximately 0.67–0.69 for models conditioned on static environmental or time-series features, while a model with no static features at all achieves a mean NSE of 0.73, outperforming all static-feature variants. Permutation importance shows that dynamic meteorological inputs matter by orders of magnitude more than any static feature. Replacing all static features with random integers gives indistinguishable in-sample performance, which shows the model uses static features as well IDs to memorise each well’s behaviour rather than learning patterns that transfer across locations. This is not specific to groundwater: Lim et al. (2021) report the same for TFT across commercial forecasting datasets, where the unique ID for each time series consistently gets the highest importance score above all other static features. Static attributes work in the rainfall-runoff literature, but do not seem to help for groundwater here, most likely because 108 wells is too few and too similar for any real correlations between static attributes and dynamics to show up. This is one reason this thesis adds a GP interpolation stage: static attributes alone are not enough to generalize to unmonitored locations. Willard, Varadharajan, et al. (2025) mark the role of static site characteristics in ML models for unmonitored prediction as an open question beyond groundwater, which means the Heudorfer finding is not a groundwater-specific exception but part of a wider unresolved problem.

GWL dynamics differ from surface-water systems in two important ways. First, groundwater responds slowly to precipitation: the subsurface introduces lags of weeks to months between precipitation events and aquifer response, which makes long look-back windows essential. The one-year look-back window used in this thesis ($L = 52$ weeks) matches this delay. Second, the spatial structure of GWL is shaped by geological heterogeneity, which cannot be inferred from time series alone. This thesis handles it by encoding well-specific hydrogeological attributes into the GRU’s initial hidden state through a linear projection layer.

Ohmer and Liesch (2026) find that global LSTM models match per-well models in accuracy across approximately 3,000 German monitoring wells, with accuracy depending more on training-well similarity to the target than on training-set size. This suggests the benefit of global training depends on how representative the training population is, a caveat relevant to this thesis.

The broader German groundwater monitoring context is documented in Ohmer, Liesch, et al. (2026), who compile a 32-year benchmark dataset (1991–2022) of 3,207 wells across Germany with meteorological inputs and over 50 site-specific static attributes. The Brandenburg LfU network used in this thesis is one component of this larger national infrastructure; this dataset could be used to test the GIST pipeline across all of Germany.

2.3. Prediction at Unmonitored and Ungauged Locations

Parameter regionalization approaches estimate model parameters for ungauged sites by using parameters of nearby gauged ones. Deep learning replaces this step with static-attribute conditioning: Kratzert, Klotz, Herrnegger, et al. (2019) show that a global model conditioned on site attributes can generalize to unseen sites without per-site training. This still assumes that static attributes are informative enough to characterize a new location’s dynamics. For groundwater, this assumption can fail when the geological setting of the target wells is not well represented in the training data, which is one of the main challenges this thesis deals with.

Li et al. (2022) extend the PUB approach by replacing static site attributes with randomly assigned embedding vectors and show that a global LSTM can represent site-specific behavior at gauged sites without being given any physical information about the site. Their finding applies only to sites seen during training: random vectors carry no information about unseen locations and cannot support prediction at genuinely unmonitored wells.

Willard, Varadharajan, et al. (2025) survey ML techniques for time series prediction at unmonitored sites across streamflow, water quality, and lake temperature. Almost all reviewed studies address streamflow or water quality; groundwater appears only marginally, and Kunz et al. (2025) is one of the few large-scale studies applying global deep learning to GWL prediction.

Varadharajan et al. (2022) discuss ML applications and opportunities for river water quality prediction and identify spatial generalization to unmonitored locations as a key open challenge. Heudorfer et al. (2024) measure the same problem for groundwater models directly (see Section 2.2). Most of these studies use leave-one-out cross-validation, which retains close spatial neighbors in training and makes the task easier than realistic deployment; the spatial split design used here addresses this (Section 4.2.3).

2.4. Spatial Interpolation of Groundwater and Geostatistical Methods

Spatial interpolation of GWL has a long history in hydrogeology, mainly through geostatistical methods. Ordinary kriging treats GWL as a spatially correlated random field (one where the correlation between any two measurement locations depends on their distance) and estimates the covariance structure from the empirical variogram, a function describing how correlation decays with distance (Rasmussen and Williams, 2006). Spatiotemporal extensions of kriging model GWL jointly over space and time. Ruybal et al. (2019) show that spatiotemporal regression kriging outperforms ordinary spatial kriging for the sparsely sampled Arapahoe aquifer in Colorado: adding temporal correlation reduces prediction error at unmonitored times. Kazemi et al. (2021) apply spatiotemporal regression kriging to the Harvey River Catchment in Western Australia and assess accuracy via leave-one-out cross-validation. J  nez-Ferreira et al. (2023) do the same for the Basin of Mexico aquifer and confirm that spatiotemporal interpolation yields smaller errors than independent spatial snapshots. In all three cases, leave-one-out cross-validation retains

close spatial neighbors in training, which makes the task easier than it would be in a realistic deployment scenario (Roberts et al., 2017) (see also Section 2.3).

Several recent studies combine machine learning with geostatistical methods. Zowam and Milewski (2024) combine ML point predictions with empirical Bayesian kriging (a variant of kriging that estimates the variogram locally around each prediction point rather than fitting one global model) for predicting GWL changes across Arizona. Like this thesis, they evaluate on well locations that are excluded from training, rather than using the same locations at different time steps. The spatial test set is small (7 out of 59 wells), however, which limits how reliable the performance estimates are.

Pradhan et al. (2024) combine a deep neural network with a GP for GWL interpolation in California’s Central Valley in an architecture similar to this thesis: the network learns a feature representation from lithological and hydrogeological inputs, and the GP interpolates in that space. Their comparison of a coordinate-only GP against a feature-augmented GP mirrors the feature ablation in this thesis. The main difference is that Pradhan et al. estimate fixed parameters of a simple trend model from past observations, without a temporal forecasting component. This thesis adds a GRU as the first stage and interpolates GRU predictions rather than observations.

GP regression is a Bayesian framework for spatial interpolation that gives both predictions and uncertainty estimates (Rasmussen and Williams, 2006). A GP with a stationary kernel is mathematically equivalent to ordinary kriging: both estimate a spatial field using a covariance function that depends only on distance (Rasmussen and Williams, 2006). The Matérn family of kernels is the standard choice in spatial statistics because it directly controls how smooth the predicted surface is; the Matérn- $\frac{3}{2}$ kernel assumes once-differentiable processes and is common for physical spatial fields (Rasmussen and Williams, 2006). GP inputs can also include spatial features beyond coordinates (covariate-augmented kriging), and which features help is not obvious in advance.

2.5. Joint Spatiotemporal Models and Neural Processes

Several models learn spatial and temporal patterns jointly in a single stage. This thesis also evaluates a jointly trained spatiotemporal model alongside the two-stage design.

Deep kernel learning (Wilson et al., 2016) replaces the fixed GP kernel with a neural network that maps inputs to a learned feature space, which gives the GP more flexibility to capture complex spatial patterns.

Neural Processes (Garnelo et al., 2018; Dubois et al., 2020) encode a set of observations into a latent representation and predict at arbitrary query locations; in effect, they learn a distribution over functions and can generalize to new contexts without retraining. Convolutional Conditional Neural Processes (Gordon et al., 2020) extend this to spatial fields. These are a direct alternative to the GRU-GP pipeline used here, but not yet applied to multi-well groundwater settings.

The jointly trained spatiotemporal model in this thesis passes GRU and GP gradients through a combined loss simultaneously; results are discussed in Chapter 5.

2.6. Uncertainty Quantification and Ensemble Methods

Uncertainty estimates matter in hydrology because forecasts influence decisions about water management and drought response. GP regression gives uncertainty estimates alongside predictions, which is useful here. Willard, Varadharajan, et al. (2025) list it among the methods for uncertainty at unmonitored locations. The GP in this thesis serves both roles: spatial interpolation and uncertainty estimation. But GP uncertainty can be poorly calibrated when predicting far from training data, especially if the kernel is fitted on data with a different spatial structure than the test set.

2.7. Reproducibility and Variance in Deep Learning

Deep learning results depend on random seed choice, which is a known issue for benchmark reliability. Bouthillier et al. (2021) show that variance from data splits, hyperparameter choices, and weight initialization can each be as large as the performance differences between models; data splits and HPO are the two largest sources, while weight initialization matters less. Willard and Varadharajan (2025) reach the same conclusion in hydrology: for stream temperature prediction, meaningful ensemble diversity comes from combining different architectures and hyperparameter configurations, not from varying random weight initialization. In spatial interpolation, which wells are in the test set adds variance, as the multi-seed experiments in this thesis show.

Bouthillier et al.’s main recommendation is to evaluate multiple random data splits, not just multiple random initializations. The multi-seed analysis in this thesis does exactly this: what changes across seeds is the spatial split: both the test set and training set composition change, and consequently so do the learned model weights. Without the $\sim 90\times$ speedup of the GRU over TFT, this would not have been feasible.

2.8. The Direct Predecessor: Kunz et al. (2025)

The work most directly related to this thesis is Kunz et al. (2025), whose approach this thesis partially replicates and extends on the Brandenburg dataset. Kunz et al. train deep learning models for seasonal GWL prediction across thousands of monitoring wells in multiple German federal states, including Brandenburg, using a single shared model. They evaluate two architectures: the Temporal Fusion Transformer (Lim et al., 2021) and N-HiTS (Challu et al., 2023), both using static hydrogeological features as inputs. On the Brandenburg dataset, their TFT achieves a median per-well RMSE of 0.093 m (S. Kunz, personal communication, 2025).

The GRU is preferred over N-HiTS for its performance on long time series and its native support for long look-back windows.

Kunz et al. evaluate on held-out future time periods: all 5,288 wells contribute to training, and the test set consists of later observations at those same wells. No well is withheld spatially. Their reported RMSE therefore reflects temporal forecast accuracy at already-monitored locations. This thesis instead predicts at wells with no observations at all, which is a harder problem.

2.9. Chapter Summary

This chapter reviews prior work across six areas: deep learning for hydrology, groundwater level prediction, ungauged-site generalization, GP spatial interpolation, joint spatiotemporal models, and evaluation design. No existing work combines a global temporal forecasting model with a GP spatial interpolation layer for groundwater, evaluated under spatial test splits at genuinely unmonitored locations. Key prior results motivating design choices include the finding that static attributes alone are insufficient for spatial generalization to unmonitored locations (Heudorfer et al., 2024), which motivates the GP interpolation stage, and that spatial test split composition drives evaluation variance more than weight initialization (Bouthillier et al., 2021). Chapter 3 describes the Brandenburg dataset used to test this approach.

3. Data

3.1. The Brandenburg Groundwater Dataset

The Brandenburg State Office for the Environment (LfU) operates a dense groundwater-level monitoring infrastructure across the state. Its groundwater database contains more than 16,000 monitoring wells in total, including inactive stations and third-party data, while the active state groundwater-level monitoring network comprises approximately 2,100 wells (Landesamt für Umwelt Brandenburg, 2024). From this broader network, we use a quality-filtered subset of 1,040 wells with weekly GWL observations starting between 1951 and 2008 and ending in 2024. These wells are distributed across Brandenburg’s full territory but naturally do not provide continuous values over the entire state’s land area, which leaves most locations unmonitored. This leads directly to the spatial interpolation challenge that motivates this thesis.

These 1,040 wells are a Brandenburg-only subset of the 5,288-well German dataset used by Kunz et al. (2025). Kunz et al. report above-average accuracy for Brandenburg specifically, probably because porous lowland aquifers with clear seasonal dynamics are easier to model than the mix of aquifer types across Germany. (They did not publish Brandenburg-specific performance metrics but shared them with us directly; these are used as a reference in Section 5.2, though the comparison is not exact since the validation-set normalization approach differs.) This is also why Brandenburg is a good choice for the spatial interpolation experiment: a strong temporal baseline would mean poor spatiotemporal results point to the interpolation stage rather than to forecast quality. The LfU recorded the data and made it available through the Federal Institute for Geosciences and Natural Resources (BGR); the dataset spans 1951–2024 with 2,142,687 total observations.

3.1.1. Target Variable

The target variable is the groundwater level (GWL) in metres above sea level (m a.s.l.), measured weekly at each well. Only wells with consistent records starting 2008 or earlier and remaining active until mid 2024 are included, so every well in the dataset covers at least the final 16.5 years of the observation period. Kunz et al. (2025) applied gap imputation to the raw measurements before sharing the data: gaps of up to four weeks were linearly interpolated, and gaps of up to 52 weeks were filled with a Bayesian imputer (S. Kunz, personal communication, 2025).

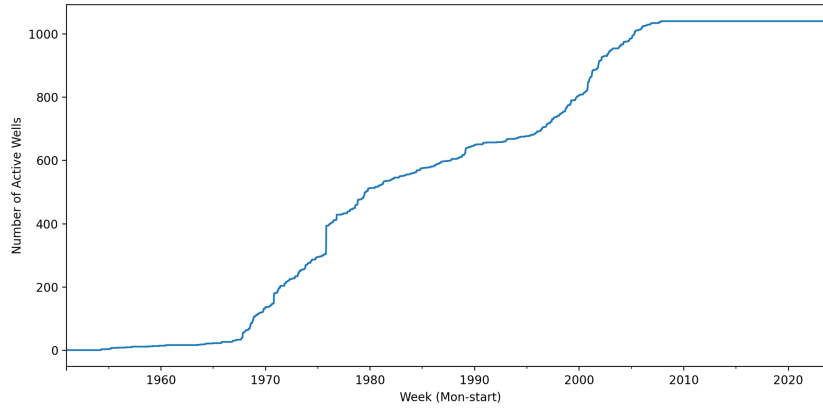


Figure 3.1.: Number of active wells per week from 1951 to 2024; network reached 1,040 active wells by 2008

Temporal coverage. Many wells were built in the mid-1970s, likely due to a major monitoring expansion during the GDR (German Democratic Republic) era. Figure 3.1 shows the growth of the active monitoring network over time, with all 1,040 wells remaining active until 2024.

Spatial coverage. The 1,040 wells are distributed across Brandenburg’s approximately 29,654 km² area, with denser coverage in the east and sparser coverage in several other parts of the state; their spatial distribution is shown in Figure 3.2. Well coordinates are provided in the EPSG:25833 (ETRS89 / UTM Zone 33N) projected coordinate system.

Because coverage is uneven, the distance from a test well to its nearest training neighbor varies widely across wells and across random split seeds. Wells in sparsely monitored parts of the state are generally harder to interpolate than wells close to a training neighbor.

3.2. Temporal and Spatial Variation

One important characteristic of the dataset is the relative magnitudes of the temporal and spatial variation of GWL.

- **Temporal variation per well:** The median within-well range of GWL is 1.57 m, with a maximum of 16.36 m.
- **Spatial variation per timestep:** The range of long-term per-well mean GWL across all active wells is approximately 130 m, which is roughly 83 times larger than the median temporal variation per well.

This large ratio of spatial to temporal variance implies that spatial context (where a well is located) is more important than temporal dynamics (when we observe a measurement) in explaining GWL values.

Figure 3.2 shows the per-well mean GWL (m a.s.l.) averaged over the full 2008–2024 observation period, which is the only period with 100% coverage. The spatial variation across the state is large: wells in low-lying areas in the north and east are close to sea level, while wells in the elevated areas of the southwest and south reach above 50 m a.s.l. To interpolate this spatial field to test well locations is the task of the GP in the second stage of the pipeline.

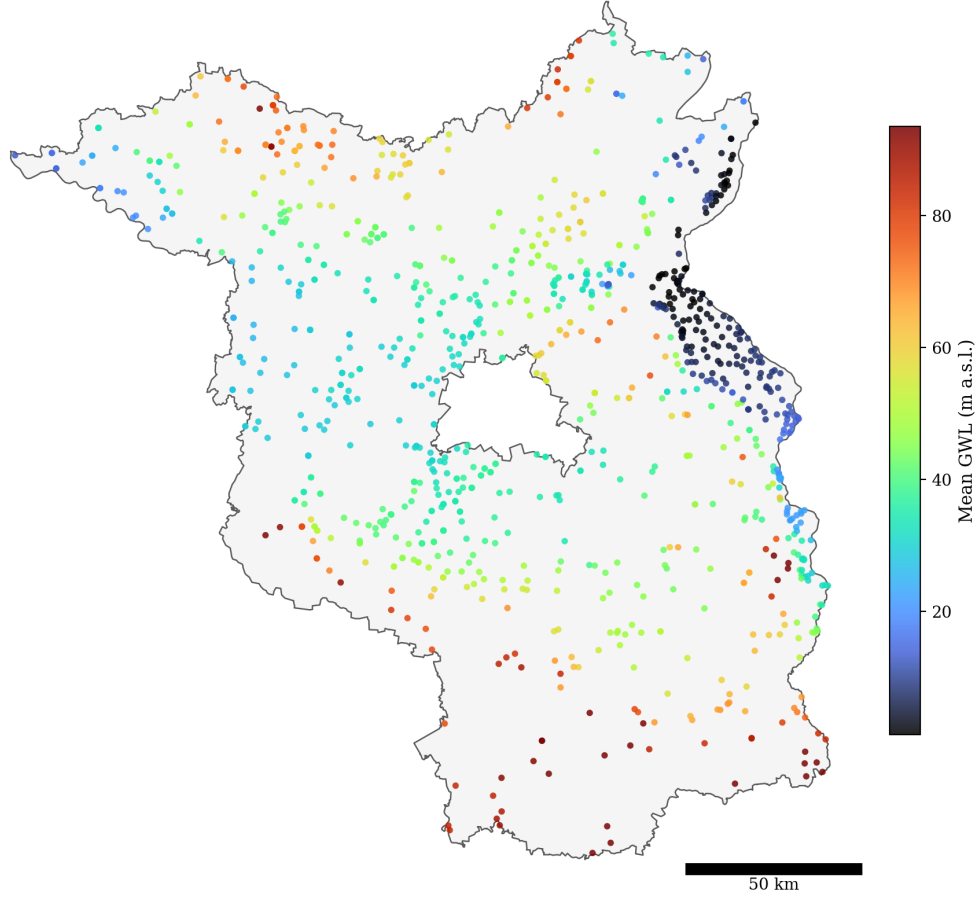


Figure 3.2.: Mean GWL per well (m a.s.l., averaged over 2008–2024) across all 1,040 monitoring wells in Brandenburg; the hole in the middle marks Berlin, which is not part of the state. The color scale is clipped at the 2nd and 98th percentiles (1.3 m and 93.6 m) to prevent extreme wells from compressing the color range; the actual range spans 0.3 m to 130.3 m.

Figure 3.3 shows observed GWL time series at three representative wells, one from each hydrogeological zone (recharge, transit, discharge). Each well is shown over a three-year window from the test period for direct comparison. Seasonal cycles are visible in all three, but the amplitude and mean level differ across zones.

3.3. Features

Each well has dynamic meteorological features and static hydrogeological features.

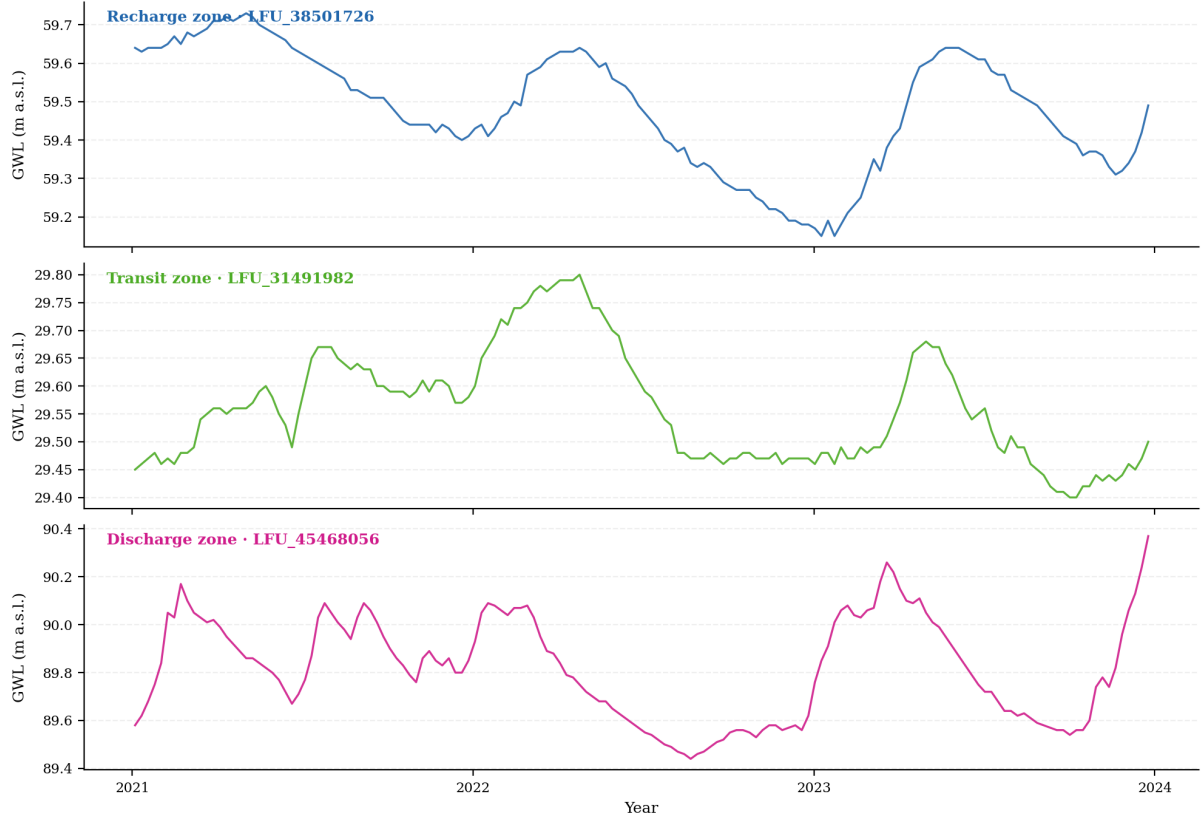


Figure 3.3.: GWL time series at three representative wells, one per hydrogeological zone

3.3.1. Dynamic Covariates

Meteorological dynamic features are taken from HYRAS 5.0 (Razafimaharo et al., 2020), a gridded meteorological dataset produced by the German Weather Service (Deutscher Wetterdienst, DWD). HYRAS 5.0 covers Germany with high-resolution meteorological grids. Daily values are aggregated to weekly resolution. Meteorological values for each monitoring well are taken from the preprocessed dataset provided by Kunz et al. (2025), aggregated within a 5 km buffer as a weighted average across intersecting grid pixels. The 5 km resolution matches that of operational DWD weather forecasts, which allows the pipeline to use forecast data in real-world deployment. The three HYRAS meteorological variables together with two calendar encodings give five dynamic features in total. The variables are:

1. **Temperature** (°C) — weekly mean of air temperature in 2 m altitude
2. **Relative humidity** (%) — weekly mean
3. **Precipitation** (mm/week) — accumulated weekly precipitation
4. **Day-of-year (sine)** — cyclic encoding $\sin(2\pi \text{DOY}/365.25)$
5. **Day-of-year (cosine)** — cyclic encoding $\cos(2\pi \text{DOY}/365.25)$

The sine and cosine encoding lets the model pick up seasonal patterns and avoids jumps when the year changes from December to January.

3.3.2. Static Features

The dataset contains 40 static hydrogeological features per well, of which 35 are used as GRU inputs; 3 of those (the one-hot encoding of hydrogeological zone into recharge, transit, and discharge categories) are also used as GP spatial inputs. The remaining 5 features are available in the dataset but are not primary model inputs. Complete feature definitions, column names, and model assignments are in Appendix C.

Three features are highlighted here because they appear prominently in the results.

Hydrogeological zone (`hydroraum`) classifies each well as sitting in a recharge zone (where rain infiltrates into the ground), a discharge zone (where groundwater flows out near streams), or a transit zone in between. Wells in different zones behave differently even under the same weather conditions (3 categories; 109, 528, and 403 wells respectively; Figure 3.4). The GP feature ablation (Section 5.3) identifies this as the single most informative static covariate for spatial interpolation.

Aquifer confinement (`gw_gespannt_bin`) records whether the aquifer is confined or unconfined; 319 confined and 721 unconfined wells (Figure 3.5).

Groundwater residence time (`siwa_verweilzeit_j`) records how long water typically stays underground before surfacing again, in years (categories: 0, 1, 3, 10, 30, or 50; 43 wells with missing values; Figure 3.6).

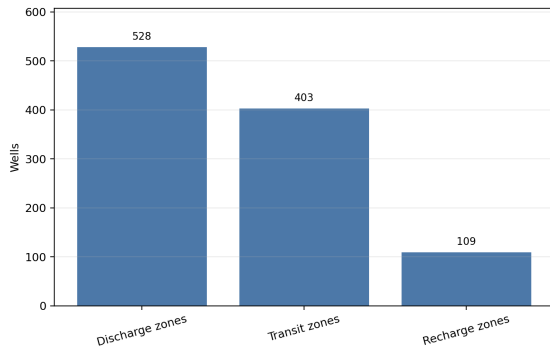


Figure 3.4.: Hydrogeological zone distribution across the 1,040 monitoring wells

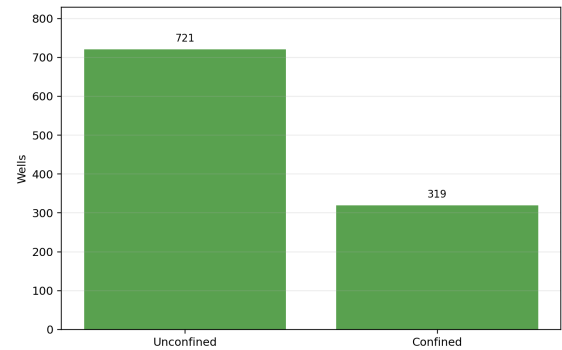


Figure 3.5.: Aquifer confinement distribution across the 1,040 monitoring wells

3.4. Data Preprocessing

The following preprocessing steps are applied, with steps 1–3 performed by Kunz et al. (2025) and steps 4–5 applied in this work:

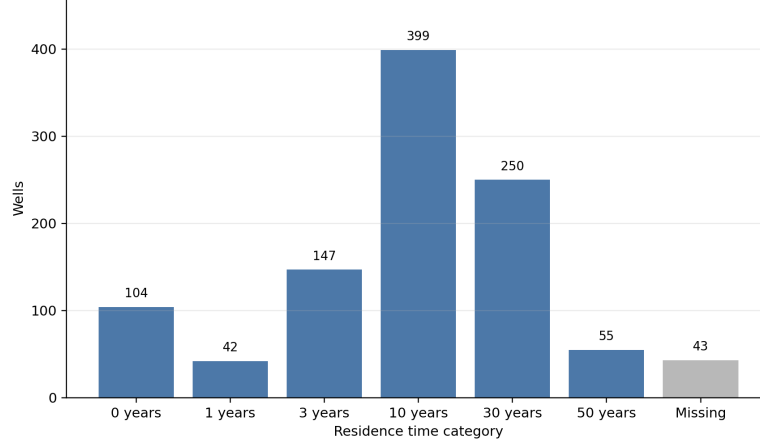


Figure 3.6.: Groundwater residence time distribution

1. **Temporal alignment:** All time series are resampled to ISO-week granularity (Monday-start weeks).
2. **Gap imputation:** Gaps of up to 4 weeks are linearly interpolated; gaps of up to 52 weeks are filled with a Bayesian imputer.
3. **Well selection:** Only wells with consistent GWL time series starting no later than 2008 and having continuous values until 2024 (after the imputations in step 2) are included.
4. **GWL normalization:** GWL values are z-scored per well (zero mean and unit variance), using the mean and standard deviation computed on the training portion of each well’s time series. This removes each well’s individual long-term mean level, so the temporal model learns dynamics (deviations from the mean) rather than absolute levels. See Section 3.2 for the motivation.
5. **Feature normalization:** Continuous static and dynamic covariates (meteorological variables and static features) are standardized per feature, again using statistics computed on the training set only.

3.4.1. Co-located Well Handling

Several wells in the dataset share essentially identical UTM coordinates (EPSG:25833) with at least one other well. This is the case at monitoring sites where multiple boreholes or screened intervals were installed at the same surface point, which is a common practice at sites requiring monitoring of different aquifer layers simultaneously.

This issue becomes clear during initial spatial evaluation experiments: the spatial GP pipeline performs unusually well on certain test wells, in some cases even outperforming the pure temporal GRU baseline that has been trained on past observations of those exact wells. This is highly unexpected, since the GP interpolation to unmonitored locations should perform substantially

worse than a model trained directly on the well’s own history. Further investigation shows that these high-performing test wells have training-set neighbors at identical or near-identical surface coordinates. Therefore, the GP effectively produces the prediction for the unmonitored location by just copying the value of the co-located training well, not by generalizing or interpolating spatially in any meaningful sense. This skews overall performance metrics, since they are dominated (and unfairly improved) by those trivially easy cases rather than by the model’s ability to interpolate to genuinely unseen locations.

On the other hand, co-located boreholes can also hurt performance when their GWL values differ substantially. In that case, the GP still copies the value from the co-located training well (since the coordinates are identical), but the copied value can be a poor match for a well in a completely different depth, which makes performance worse rather than better. Since there is no clean depth threshold that separates the cases where interpolation is too easy from the ones where it is too hard, the cleanest solution is to put all co-located wells into the training set. This makes sure all measurements are kept and we are not discarding valuable data.

More specifically, the split generation algorithm identifies all wells within 8 m of at least one other well (308 wells in total) and puts them permanently into the training set. These wells are never eligible to appear in the test set. This approach keeps all 1,040 wells in the dataset, so no measurements are discarded, while guaranteeing that every well in the test set poses a genuine spatial interpolation challenge. Preliminary experiments show that excluding wells by aquifer type hurts performance, suggesting that having more training data matters more than having perfectly clean data.

The remaining 732 wells make up the eligible test pool from which the random 90/10 split is drawn (splitting all 1,040 wells randomly would not give meaningful results, because of the co-located wells). This yields 104 test wells and 936 training wells per split seed.

3.5. Chapter Summary

This chapter describes the Brandenburg groundwater dataset: 1,040 wells with weekly GWL observations starting between 1951 and 2008 and ending in 2024, five dynamic meteorological features from HYRAS 5.0, and 40 static hydrogeological features per well (of which 35 are used as GRU inputs; see Appendix C). The large ratio of spatial to temporal GWL variation (roughly 83:1 in median ranges) motivates the two-stage pipeline design (Section 4.2.1), and the 308 co-located wells are forced into training to prevent trivially easy test cases. The uneven spatial distribution of wells means test wells in different split seeds vary widely in their isolation from the nearest training neighbor. Chapter 4 presents the full methodological framework built on this dataset.

4. Methods

This chapter presents the full methodological framework in three parts. Section 4.1 provides the technical background on Gated Recurrent Units, Gaussian Process regression, and the evaluation metrics used throughout. Section 4.2 describes the design of the temporal and spatial models, the pipeline variants, and the spatial split. Section 4.3 documents the software setup and compute infrastructure.

4.1. Technical Background

4.1.1. Recurrent Neural Networks and the Gated Recurrent Unit

Sequence-to-Sequence Forecasting

A sequence-to-sequence (seq2seq) model maps an input sequence of length L to an output sequence of length H . Given a window of L historical observations and input features $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_L)$, the model predicts the next H values $\hat{\mathbf{y}} = (\hat{y}_{t+1}, \dots, \hat{y}_{t+H})$. This study uses $L = 52$ weeks (one year of look-back) and $H = 16$ weeks (approximately four months of forecast horizon).

Gated Recurrent Unit

The Gated Recurrent Unit (Cho et al., 2014) is a recurrent neural network architecture that addresses the vanishing gradient problem of standard RNNs through learnable gating mechanisms. Given input $\mathbf{x}_t$ and previous hidden state $\mathbf{h}_{t-1}$, the GRU computes:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \quad (4.1)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \quad (4.2)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad (4.3)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (4.4)$$

where σ is the sigmoid function, $\odot$ stands for element-wise multiplication, $\mathbf{z}_t$ is the update gate, $\mathbf{r}_t$ is the reset gate, $\tilde{\mathbf{h}}_t$ is the candidate new hidden state, $\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}_h$ are learned input weight matrices, $\mathbf{U}_z, \mathbf{U}_r, \mathbf{U}_h$ are learned hidden weight matrices, and $\mathbf{b}_z, \mathbf{b}_r, \mathbf{b}_h$ are bias vectors. The

update gate controls how much of the previous hidden state is retained; the reset gate controls how much of the past is forgotten when computing the candidate hidden state.

Static Feature Integration

Well-specific static features (hydrogeological covariates that are constant over time) are integrated here by projecting them into the initial hidden state $\mathbf{h}_0$ via a learned linear layer:

$$\mathbf{h}_0^{(\ell)} = \mathbf{W}_s^{(\ell)} \mathbf{s} + \mathbf{b}_s^{(\ell)}, \quad \ell = 1, \dots, L_{\text{layers}} \quad (4.5)$$

where $\mathbf{s} \in \mathbb{R}^{d_s}$ is the static feature vector of dimension d_s , $\mathbf{W}_s^{(\ell)}$ and $\mathbf{b}_s^{(\ell)}$ are the learned weight matrix and bias for layer ℓ , and L_{layers} is the total number of GRU layers. This allows the same model weights to be shared across all wells while the well-specific context is encoded in the initial hidden state.

Comparison to the Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) (Lim et al., 2021) is an attention-based seq2seq architecture designed for multi-horizon forecasting with heterogeneous inputs. It incorporates variable selection networks, gating mechanisms, and multi-head attention over the entire input sequence. While TFT achieves strong performance on such tasks, it carries significantly more parameters and longer training times than a GRU. The GRU is therefore selected as the temporal model; a replication of the Kunz et al. (2025) TFT baseline and a direct performance comparison are presented in Section 5.2.1. The GRU in this thesis is trained with MSE loss, which is standard for GRU regression models. RMSE is used as the evaluation metric for both models, so their values are directly comparable regardless of training loss.

4.1.2. Gaussian Process Regression

Definition

A Gaussian Process is a distribution over functions $f : \mathcal{X} \rightarrow \mathbb{R}$ such that any finite collection of function values is jointly Gaussian (Rasmussen and Williams, 2006):

$$f(\mathbf{x}_1), \dots, f(\mathbf{x}_n) \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K}) \quad (4.6)$$

where $\mathbf{x}_i \in \mathcal{X}$ is the input point of the i -th well, $\boldsymbol{\mu}$ is the mean vector with $\mu_i = m(\mathbf{x}_i)$, and $\mathbf{K}$ is the covariance matrix with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$. The mean function $m(\cdot)$ and kernel $k(\cdot, \cdot)$ define what the GP assumes about the function before seeing any data.

Posterior Prediction

Given training observations $\mathbf{y} = f(\mathbf{X}) + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, the GP posterior at test points $\mathbf{X}_*$ is:

$$\mathbf{f}_* \mid \mathbf{X}, \mathbf{y}, \mathbf{X}_* \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*) \quad (4.7)$$

$$\boldsymbol{\mu}_* = m(\mathbf{X}_*) + \mathbf{K}_* \left(\mathbf{K} + \sigma_n^2 \mathbf{I} \right)^{-1} (\mathbf{y} - m(\mathbf{X})) \quad (4.8)$$

$$\boldsymbol{\Sigma}_* = \mathbf{K}_{**} - \mathbf{K}_* \left(\mathbf{K} + \sigma_n^2 \mathbf{I} \right)^{-1} \mathbf{K}_*^\top \quad (4.9)$$

where $\mathbf{K} = k(\mathbf{X}, \mathbf{X})$ is the training kernel matrix, $\mathbf{K}_* = k(\mathbf{X}_*, \mathbf{X})$ is the cross-covariance between test and training points, and $\mathbf{K}_{**} = k(\mathbf{X}_*, \mathbf{X}_*)$ is the test kernel matrix, and σ_n^2 is the observation noise variance. The posterior mean $\boldsymbol{\mu}_*$ provides point predictions, while the diagonal of $\boldsymbol{\Sigma}_*$ provides predictive variances.

The Matérn-3/2 Kernel

Matérn kernels control the smoothness of the modelled function. The Matérn-3/2 kernel is:

$$k_{\text{M32}}(r) = \sigma^2 \left(1 + \frac{\sqrt{3} r}{\ell} \right) \exp \left(-\frac{\sqrt{3} r}{\ell} \right) \quad (4.10)$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$ is the Euclidean distance between two wells, ℓ is the length-scale (controlling how quickly spatial similarity decays with distance), and σ^2 is the output-scale (the overall variance of the GP). Matérn-3/2 assumes once-differentiable sample paths, meaning each function the GP contains has exactly one continuous derivative, which is a standard prior for spatially smooth but not infinitely smooth physical fields such as GWL.

Kernel Hyperparameter Optimisation

The kernel hyperparameters, length-scales ℓ , output scale σ^2 , and noise variance σ_n^2 , are not set by hand but learned from data by maximising the marginal log-likelihood (MLL). The MLL is the log probability of the observed targets $\mathbf{y}$ under the GP model, integrated over all possible latent function values. It balances data fit against model complexity, and its gradient with respect to the hyperparameters is available in closed form (Rasmussen and Williams, 2006), which allows gradient-based optimisation.

Exact Inference with GPyTorch

Exact GP inference requires inverting the $n \times n$ kernel matrix, where n is the number of training points, which has $\mathcal{O}(n^3)$ time complexity and $\mathcal{O}(n^2)$ memory complexity (Rasmussen and Williams, 2006). Since there are at most 936 training wells per date in our setup, exact inference is computationally feasible. This thesis uses the GPyTorch library (Gardner et al., 2018) for its kernel and likelihood implementations, but solves the GP equations exactly using Cholesky decomposition instead of GPyTorch’s built-in approximate solvers, because $n \leq 936$ makes exact inference feasible and Cholesky gives the exact solution with no approximation error. Rather than computing $(\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1}$ directly, Cholesky factorises $\mathbf{K} + \sigma_n^2 \mathbf{I} = \mathbf{L}\mathbf{L}^\top$, where $\mathbf{L}$ is a lower-triangular matrix. The posterior mean and variance then follow from two triangular solves rather than an explicit inversion, which is faster and numerically more stable. At $n \approx 936$, the Cholesky factorization involves roughly 10^8 floating-point operations per fit, which completes in milliseconds on the GPU hardware used (see Section 4.3.2).

4.1.3. Evaluation Metrics

Results are reported primarily at forecast horizon $h = 16$ (16 weeks ahead), which is the most demanding horizon and therefore the most informative single value to report. The choice also allows comparability with unpublished experiments on the Brandenburg dataset shared by Kunz et al.; their published paper uses $h = 12$, but on a different dataset, so those published values are not directly comparable regardless. The mean of each metric over all 16 horizons is additionally reported. Each per-well value at a given horizon is averaged over all evaluation time steps T_i ; distribution statistics are then computed across test wells.

Root Mean Squared Error

The per-well RMSE at forecast horizon h is:

$$\text{RMSE}_i^h = \sqrt{\frac{1}{|T_i|} \sum_t (\hat{y}_{i,t+h} - y_{i,t+h})^2} \quad (4.11)$$

where i indexes the test well, t indexes the evaluation time steps in T_i , $\hat{y}_{i,t+h}$ is the model prediction, and $y_{i,t+h}$ is the observed GWL value. The primary metric is RMSE_i^{16} ; the mean $\overline{\text{RMSE}}_i = \frac{1}{H} \sum_{h=1}^H \text{RMSE}_i^h$ is additionally reported.

The following distributional summaries are reported to characterize both central tendency and tail behavior: the mean $\pm$ standard deviation across test wells, and the percentiles P10, P25, median (P50), P75, and P90 of the per-well RMSE distribution (Table B.9). P10 marks the best-performing 10% of wells; P90 is the threshold above which only the worst-performing 10% fall; P25–P75 (interquartile range) shows how concentrated results are around the median.

Normalized RMSE per Well (nRMSE_{pw})

Because GWL ranges vary across wells, raw RMSE is larger for wells with bigger amplitudes even at equal model accuracy; normalizing by a well-specific scale removes this and gives each well equal weight:

$$\text{nRMSE}_{\text{pw},i}^{\text{IQR}} = \frac{\text{RMSE}_i}{\text{IQR}(y_{i,\cdot})} \quad (4.12)$$

where $\text{IQR}(y_{i,\cdot}) = Q_{75}(y_{i,\cdot}) - Q_{25}(y_{i,\cdot})$ is the interquartile range of observed GWL values at well i . The result is aggregated as the median across test wells. Range normalization, $\text{RMSE}_i/\text{range}(y_{i,\cdot})$ where $\text{range}(y_{i,\cdot}) = \max(y_{i,\cdot}) - \min(y_{i,\cdot})$, is a natural alternative, but across the wells in this study the range is about 3–4 $\times$ larger than the IQR and more sensitive to extreme observations. IQR normalization is therefore more robust and is the single reported normalized metric. It is used for pipeline comparison and for the GP feature ablation (Section 5.3).

Nash–Sutcliffe Efficiency

The Nash–Sutcliffe Efficiency (NSE) (Nash and Sutcliffe, 1970) measures the fraction of variance explained relative to a climatological mean baseline:

$$\text{NSE}_i^h = 1 - \frac{\sum_t (\hat{y}_{i,t+h} - y_{i,t+h})^2}{\sum_t (\bar{y}_i - y_{i,t+h})^2} \quad (4.13)$$

where $i, t, h, \hat{y}_{i,t+h}, y_{i,t+h}$ are as in Equation (4.11), and $\bar{y}_i$ is the per-well mean GWL baseline. Two choices exist for $\bar{y}$: the test-period mean or the training-period mean. This study uses the test-period mean as the primary choice, consistent with the original formulation of Nash and Sutcliffe (1970), who define $\bar{y}$ as the mean of observed values over the evaluation period, the standard in hydrology. The primary metric is NSE_i^{16} ; the mean $\overline{\text{NSE}}_i = \frac{1}{H} \sum_{h=1}^H \text{NSE}_i^h$ is additionally reported. $\text{NSE} \in (-\infty, 1]$: a value of 1 indicates perfect prediction, 0 indicates no better than the mean, and negative values indicate worse-than-mean performance.

NSE is reported for all models. For the temporal models it is a standard performance measure; for the spatiotemporal pipeline, the structural disadvantage means values will be strongly negative and should be interpreted accordingly. For spatial test wells, NSE has a structural disadvantage compared to the temporal models: its baseline uses the per-well mean $\bar{y}$, which is derived from observations at that specific well, which is information the GP never has access to during training. Both the temporal models and the GP use the per-well test-period mean as the NSE baseline, which neither model has seen during training. The disadvantage is larger for spatial interpolation, though: the GP has to estimate GWL levels per timestep at unobserved wells across a range of roughly 130 m, while each temporal model works within a single well whose amplitude over the entire time span is on average around 1.5 m and has a maximum value of 16 m. Missing

the absolute level by even a small fraction of the range produces a much larger NSE penalty than a comparable relative miss within a single well. As a result, spatiotemporal pipeline NSE values are strongly negative, regardless of pipeline variant (mean ≈ -38.4 for the primary 90/10 evaluation) and are reported in the results.

Kling-Gupta Efficiency

The Kling-Gupta Efficiency (Gupta et al., 2009) separates prediction error into three components (correlation, variability, and bias), which gives a more complete picture than NSE alone:

$$\text{KGE} = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2} \quad (4.14)$$

where r is the Pearson correlation coefficient between predicted and observed values, $\alpha = \sigma_s/\sigma_o$ is the ratio of simulated to observed standard deviation (variability ratio), and $\beta = \mu_s/\mu_o$ is the ratio of simulated to observed mean (bias ratio). A perfect model achieves $\text{KGE} = 1$; values closer to 1 are better. Unlike NSE, always predicting the observed mean does not correspond to $\text{KGE} = 0$: it typically gives $\text{KGE} \approx -0.41$ (Knoben et al., 2019), so $\text{KGE} = 0$ already represents a model that outperforms that baseline. Unlike NSE, which is based on squared residuals alone, KGE explicitly accounts for correlation, variability, and bias in how it penalizes prediction errors. Per-well KGE is computed using the same predicted and observed values as RMSE and NSE.

Global (Pooled) Metrics

Per-well summary statistics treat each well as an independent observation and are robust to outliers. Complementarily, global metrics pool all errors across all test wells $i \in \mathcal{W}_{\text{test}}$, all evaluation time steps t , and all horizons h :

$$\text{RMSE}_{\text{global}} = \sqrt{\frac{1}{|\mathcal{W}_{\text{test}}| \cdot |T| \cdot H} \sum_{i,t,h} (\hat{y}_{i,t+h} - y_{i,t+h})^2} \quad (4.15)$$

$$\text{NSE}_{\text{global}} = 1 - \frac{\sum_{i,t,h} (\hat{y}_{i,t+h} - y_{i,t+h})^2}{\sum_{i,t,h} (\bar{y} - y_{i,t+h})^2} \quad (4.16)$$

where $\mathcal{W}_{\text{test}}$ is the set of test wells, $|T|$ is the number of evaluation time steps, $H = 16$ is the forecast horizon length, $\bar{y}$ is the global mean over all pooled observations, and $i, t, h, \hat{y}_{i,t+h}, y_{i,t+h}$ are as defined in Equation (4.11). Global RMSE gives more weight to the worst-performing wells (squaring errors amplifies large deviations, so a few badly-predicted wells raise it noticeably), so it picks up issues the per-well median can miss. Global NSE uses the grand mean across all pooled observations as its baseline, so a model that predicts the grand mean everywhere scores 0. A model only needs to place each well’s mean on the correct side of the grand mean to beat

this baseline; with a range of roughly 130 m across Brandenburg wells, that is not a hard task. The GP does exactly this, since spatial interpolation of between-well level differences is its core task. Per-well NSE uses each well’s own test-period mean as the baseline. The temporal range within a single well averages roughly 1.5 m, so the per-well mean is at most about 0.75 m from any observation in an average well. To score above zero on per-well NSE, the GP would have to predict each well’s absolute GWL to within that margin, on a scale that spans 130 m across the study area, which is not achievable for a model that has never observed that well. The result is global NSE around +0.92 and per-well NSE around −33 for the spatiotemporal pipeline, and both are reported in the results.

Prediction Interval Coverage Probability

The GP spatial interpolation stage outputs a predicted GWL $\hat{y}_t$ and a posterior standard deviation $\hat{\sigma}_t$ at each test well. A $(1 - \alpha)$ prediction interval is $[\hat{y}_t - z_{\alpha/2}\hat{\sigma}_t, \hat{y}_t + z_{\alpha/2}\hat{\sigma}_t]$, where α is the tail probability and $z_{\alpha/2}$ is the standard normal quantile; for a 90% interval, $\alpha = 0.10$ and $z_{0.05} \approx 1.645$. PICP measures the fraction of true observations that fall inside:

$$\text{PICP} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i \in [\hat{y}_i - z_{\alpha/2}\hat{\sigma}_i, \hat{y}_i + z_{\alpha/2}\hat{\sigma}_i]] \quad (4.17)$$

A well-calibrated model achieves $\text{PICP} \approx 1 - \alpha$: for a 90% interval, roughly 90% of observations should fall inside. If coverage is too low the model is overconfident; if too high, the intervals are unnecessarily wide.

Statistical Tests

Two statistical tests are used to quantify the strength and significance of relationships in the multi-seed evaluation.

The **Spearman rank correlation** ρ measures the monotonic association between two variables without assuming a linear relationship or normal distribution. It is computed on ranks rather than raw values, making it robust to outliers. A value of $\rho = 1$ indicates a perfect positive monotonic relationship; $\rho = 0$ indicates no monotonic association. The associated p -value tests the null hypothesis that the two variables are uncorrelated in the population.

The **paired t -test** compares the means of two matched samples. In this thesis it is applied to the per-seed difference in RMSE between two pipeline configurations evaluated on the same 40 random splits. The test statistic is $t = \bar{d}/(s_d/\sqrt{n})$, where $\bar{d}$ is the mean pairwise difference, s_d its standard deviation, and $n = 40$ the number of split seeds. The two-sided p -value tests the null hypothesis that the true mean difference is zero.

4.1.4. Hyperparameter Optimization

Hyperparameters for both the GRU and the GP are selected by iterative grid search over predefined search spaces. Each round evaluates all combinations in the grid, identifies the best-performing configuration, and uses it to define a narrower search space for the next round. Four rounds are run for the GRU and three for the GP. GRU configurations are evaluated on a temporal validation set (2016–2019); GP configurations are evaluated on spatial validation wells.

4.2. System Design

4.2.1. Overview

The GIST system addresses two coupled prediction tasks: temporal forecasting and spatial interpolation. The high-level design decomposes the problem into two stages, which can be executed independently (two-stage spatiotemporal pipeline) or trained jointly (jointly trained spatiotemporal model). A third configuration replaces GRU outputs with true observations to isolate the contribution of each stage (true-observation interpolation baseline).

The large spatial-to-temporal variance ratio in the dataset (Section 3.2) means a model needs to separate the long-term mean level at each location from the temporal dynamics. Per-well normalization handles this by subtracting each well’s long-term mean level and dividing by its standard deviation before training, so the model learns deviations rather than absolute groundwater levels. This decomposition into a spatial per-well mean and temporal deviations motivates the two-stage pipeline: the GRU learns temporal dynamics on normalized series, after which the GP reconstructs the spatial mean at unobserved locations. Kunz et al. (2025) apply no per-well normalization; as a result, the standard deviation of GWL is their single most important static feature.

Beyond predictive accuracy, the evaluation includes explicit interpretability analyses. A GRU feature permutation importance study and a systematic GP feature ablation, both in Chapter 5, show which input features the GRU relies on for temporal forecasting and which spatial features the GP relies on for interpolation.

Figure 4.1 shows the three model configurations. In all cases, the final output is a 16-week GWL forecast for each of the test wells.

4.2.2. Temporal Split

All 1,040 wells are assigned to one of three temporal periods. The **training period** covers each well’s available data from its start of coverage (between 1951 and 2008) through the end of 2015 and is the only period on which GRU weights are updated. The **validation period**

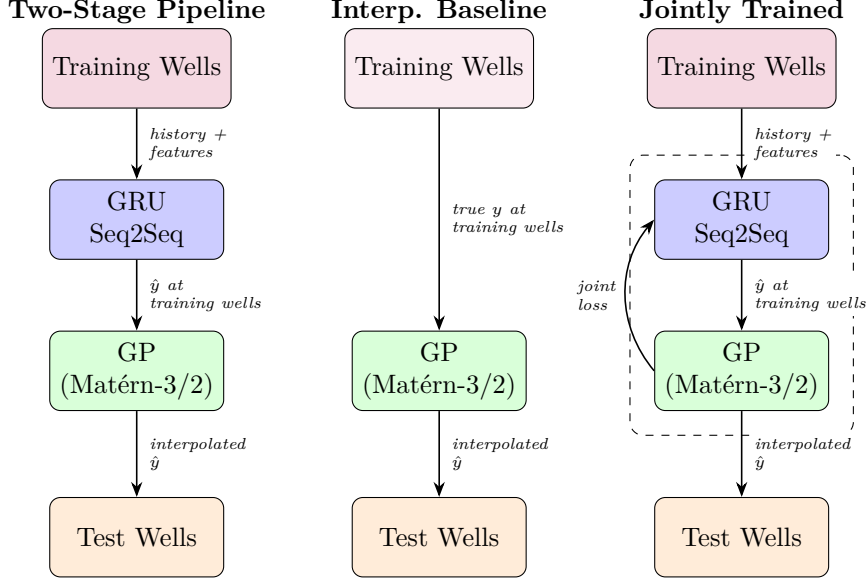


Figure 4.1.: The three model configurations evaluated in this thesis (y : true observed GWL; $\hat{y}$: model prediction). In the two-stage spatiotemporal pipeline (left), the GRU and GP are trained independently: the GRU forecasts GWL at training wells, and the GP spatially interpolates those forecasts to test wells. The true-observation interpolation baseline (centre) replaces GRU forecasts with true observations, giving an upper bound on what the two-stage pipeline can achieve if the temporal forecasts are perfect. In the jointly trained spatiotemporal model (right), GRU and GP are trained end-to-end with a joint loss.

covers 2016 through the end of 2019 and is used for GRU hyperparameter optimisation and early stopping. The **test period** covers 2020 through mid-2024 and is held out until final evaluation. All spatiotemporal forecasts reported in this thesis are evaluated on test-period predictions at spatial test wells that are never seen during training.

4.2.3. Spatial Test Split

Co-location Constraint

Before the split is drawn, all wells within 8m of at least one other well are permanently assigned to training (see Section 3.4.1). This leaves 732 eligible test wells.

Random Split

Wells are randomly assigned to train and test sets. The primary evaluation uses a 90/10 split on the full 1,040-well dataset under the co-location constraint, giving 936 training wells and 104 test wells with independent randomisation across 40 different split seeds. In a random split, test wells are distributed geographically among training wells (Figure 4.2).

Brandenburg has an unusually dense groundwater monitoring network (see Section 3.1). The core question of this thesis is what such a network can achieve for the forecast-anywhere problem. An 80% training fraction is not used: test-set stability is already sufficient at 104 wells per seed, and the larger training set at 90% better reflects what Brandenburg’s full monitoring network can achieve, which is the core question of this study. A 95% training fraction is also tested but dropped: with only about 52 test wells per seed, per-seed RMSE is noisy and the across-seed standard deviation is high. The 90/10 split keeps 936 training wells close to the full available network while giving 104 test wells per seed for stable aggregate statistics. The 10% final test fraction is also consistent with Kunz et al. (2025), who reserve similar fractions for temporal validation and test.

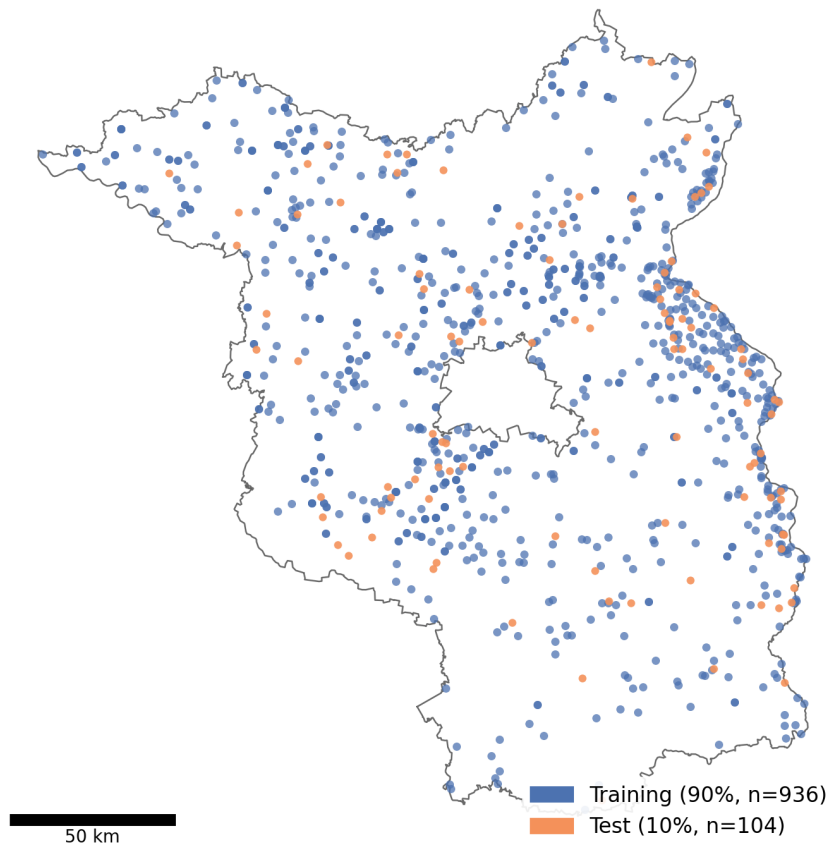


Figure 4.2.: Random 90/10 spatial test split (blue = training, orange = test). Co-located wells are forced into training. Test wells are distributed geographically across the monitoring network.

Why Leave-One-Out Cross-Validation Was Not Used

Leave-one-out cross-validation (LOOCV) is a natural candidate for spatial evaluation, but it is not considered an appropriate protocol for the research question this thesis addresses.

Conceptual mismatch. As discussed in Section 2.3, LOOCV retains each test well’s closest spatial neighbors in training, so the GP interpolates over a very short spatial gap and the measured error reflects gap-filling within a near-complete network rather than generalization to genuinely unmonitored locations. Co-located wells could in principle be included by grouping wells at the same location and holding out entire groups, but some test folds would then contain multiple test wells, which complicates the procedure.

Furthermore, the key empirical finding of this thesis (that per-well interpolation error correlates positively with distance to the nearest training neighbor) would be essentially invisible under LOOCV: by construction, most test wells have a very close training neighbor, so the error distribution would be heavily concentrated near zero distance and miss the tail behavior that is the most scientifically interesting part of the result.

Computational cost. For a global GRU model, LOOCV would require 732–1,039 separate full pipeline runs (depending on how co-located wells are handled), making this approach infeasible at this data scale: at roughly 24 minutes per two-stage pipeline run, that amounts to approximately 13–18 days of compute on a single GPU, and approximately 30–45 days for the jointly trained pipeline.

Alternative Test Strategies

Three additional test configurations are explored alongside the primary random split to assess how the choice of split design shapes reported performance. All three are evaluated as a sensitivity analysis in Section 5.7.

Dense-area split. All distances between each eligible (non-co-located) well and its three nearest neighbors are computed and ranked. The 50th percentile of this distribution identifies the wells with the shortest typical neighborhood distances, those in the denser half of the monitoring network. Test wells are then drawn randomly from this restricted pool at a 90/10 fraction. Because every test well in the Dense-area split is drawn from an area with close training neighbors, the spatial interpolation task is easier on average than under a fully random draw. The 50th percentile is a fixed design boundary, not a tuned threshold: lowering it would change which wells are in the training set, making it a different experiment rather than a sensitivity check on the same model.

Cluster-stratified split. The 732 eligible (non-co-located) wells are partitioned into $k = 20$ clusters using k -means on all 35 static hydrogeological features. A fixed fraction of approximately 14% is drawn from each cluster, giving a total close to 98 test wells; the exact count may differ by a few wells due to rounding within clusters. This distributes test wells uniformly across the static-feature space, regardless of their spatial position. Unlike the random split, which

distributes training and test wells geographically at random, this strategy ensures that every part of the hydrogeological feature space is represented in both training and test sets.

Sparse-area split. The Sparse-area split reverses the logic of the Dense-area split: test wells are drawn from the 20% of eligible (non-co-located) wells with the largest nearest-neighbor distances among eligible wells, those above the 80th percentile of that distribution; every test well therefore lies in a sparse area of the monitoring network. This creates a harder interpolation task than the random split, since the GP must generalize over larger distances with fewer nearby training wells. The two area-based splits cover opposite ends of interpolation difficulty: the Dense-area split creates an easier interpolation task, the Sparse-area split a harder one.

The area-based splits are not equivalent to post-hoc subsampling: by reassigning wells between train and test, they change the data the model trains on, so the same model cannot be evaluated under both conditions without retraining. The cluster-stratified split additionally guarantees feature-space coverage that random sampling may not provide. The sensitivity analyses therefore test generalization under genuinely different training conditions, and their consistency with the random-split results strengthens rather than merely repeats that evidence.

4.2.4. GRU Model Design

The GRU follows the seq2seq architecture described in Section 4.1.1. It consists of six stages:

- ① *Data preparation.* The 1 040 wells are split spatially into train and test sets. The temporal axis is further divided into a training, validation, and test period on the spatial train wells. GWL is normalized per well (z-score); weather covariates and static features are normalized globally. A sliding-window procedure creates overlapping 52-week input / 16-week output samples, with a 1-week shift between consecutive windows.
- ② *Model inputs.* Each sample contains $\mathbf{x}_{\text{past}}$ (52 time steps of GWL and 5 weather covariates), $\mathbf{x}_{\text{future}}$ (16 time steps of future weather covariates), $\mathbf{x}_{\text{static}}$ (35 static well attributes), and $\mathbf{y}_{\text{future}}$ (16 normalized GWL values used as training targets).
- ③ *Static projection.* The 35 static features are passed through a linear layer ($\text{Linear}(35 \rightarrow 256 \times 4 = 1\,024)$) and reshaped to provide each of the 4 encoder layers with a dedicated 256-dimensional initial hidden state, encoding the identity and characteristics of the well.
- ④ *Encoder GRU.* The encoder has 4 layers with 256 nodes each and dropout 0.2 between layers. It processes the 52-week input sequence step by step; the final hidden state of each layer is passed to the corresponding decoder layer.
- ⑤ *Decoder GRU.* The decoder has the same 4-layer, 256-node structure. It is initialized with the encoder’s final hidden state and processes 16 steps of future weather covariates as step-by-step input, autoregressively building up a representation of the forecast period.

- ⑥ *Output head, denormalization, and predictions.* A linear layer (`Linear(256 → 1)`) is applied to each of the 16 decoder outputs, producing 16 normalized GWL predictions. These are then denormalized using the per-well z-score parameters fitted on the training period, converting each value back to absolute GWL in metres above mean sea level. During training, the mean squared error between the normalized predictions and `y_future` is used as the loss.

Figure 4.3 illustrates the complete architecture across all six stages.

GRU Static Features

The GRU uses 35 static features per well, drawn from the full feature set described in Section 3.3. The raw aquifer confinement flag is replaced by its binarized equivalent (`gw_gespannt_bin`), keeping the total at 35. The hydrogeological zone is used both as a GRU input (one-hot encoded into three categories) and as the primary GP spatial feature (Section 4.2.5), where the GP feature ablation identifies it as the single most informative covariate for spatial interpolation (Section 5.3).

Best Hyperparameter Configuration

The best GRU configuration is identified through iterative grid search (Section 4.1.4) over hidden dimension, layer count, dropout, learning rate, and batch size, evaluated on a temporal validation period. The selected configuration is:

- *Architecture:* hidden dimension 256, 4 layers, dropout 0.2
- *Optimisation:* learning rate 2×10^{-4} , batch size 4096, up to 50 epochs with early stopping (patience = 5)
- *Sequence lengths:* input window 52 weeks, output horizon 16 weeks

This configuration trains in roughly 20 minutes on an NVIDIA A100 GPU and is used for all multi-seed spatiotemporal forecast interpolation experiments throughout this thesis.

4.2.5. Gaussian Process Design

The GP interpolation pipeline takes GRU-predicted GWL values at all training wells, together with their coordinates and hydrogeological zone (one-hot encoded), and assembles them per forecast horizon. For each horizon separately, an exact Gaussian Process with a Matérn-3/2 kernel is fitted. The kernel uses Automatic Relevance Determination (ARD): each input feature gets its own length-scale, so the model learns how much geographic distance and zone membership matter for spatial similarity. The fitted GP produces a GWL prediction and an uncertainty estimate at each test well.

Input Space

The GP operates on a per-timestep snapshot: at each forecast date and horizon, the vector of GWL values at all training wells is treated as a spatial field. The input $\mathbf{x}_i$ to the GP is the coordinate vector of well i , optionally augmented with static hydrogeological features. GRU predictions at training well locations serve as the target vector $\mathbf{y}$ in the GP equations (Section 4.1.2); in the true-observation interpolation baseline, true observations are used instead. The GP posterior then yields predictions at test well locations.

Mean Function

A zero mean function is used. Preliminary multi-seed experiments show a constant mean function performs worse, likely because the kernel’s output scale already captures the overall level and a constant offset adds nothing.

Kernel

The Matérn-3/2 kernel (Equation (4.10)) is applied in the augmented input space (coordinates + static features). Automatic Relevance Determination (ARD) is used: each input dimension receives its own length-scale parameter ℓ_d , allowing the kernel to weight coordinate dimensions and static features differently. The ARD length-scales are jointly optimized with the noise variance and output scale via marginal likelihood maximisation.

Kernel Pretraining and Per-Date Inference

Evaluating the GP across all test dates and forecast horizons requires fitting kernel hyperparameters in a way that is computationally tractable and consistent. The procedure used here has two levels.

Per-horizon kernel pretraining. For each of the 16 forecast horizons separately, all available training-well predictions (or true observations, for the interpolation baseline) across the entire test period are pooled into a single dataset. From this pooled dataset, 128 points are randomly subsampled and used to optimize the Matérn-3/2 kernel hyperparameters (length-scales, output scale, and noise variance) by maximising the marginal log-likelihood via 300 steps of Adam gradient descent (Kingma and Ba, 2015), an adaptive-moment optimizer that maintains per-parameter learning rates. This yields 16 sets of kernel hyperparameters, one per horizon (Appendix A.2). The pooled dataset spans the full test period and covers the full spatial distribution of training wells, so even a small subsample represents the global spatial correlation structure well; 128 points suffice for a 7-parameter kernel optimization, and the per-date Cholesky inference that follows uses all $n \approx 936$ training wells.

Per-(date, horizon) Cholesky inference. Once the kernel hyperparameters are fixed for a given horizon, the GP posterior at test well locations is computed by exact Cholesky factorisation for every (date, horizon) pair in the test period. At each such pair, the $n \times n$ kernel matrix is formed from the $n \approx 936$ training-well inputs available on that date, and the Cholesky decomposition $\mathbf{K} + \sigma_n^2 \mathbf{I} = \mathbf{L}\mathbf{L}^\top$ is solved to obtain the posterior mean and variance at each test well. Kernel hyperparameters are not updated between dates; the pretraining step already captures the global spatial structure.

This separation of kernel optimization (once per horizon, on pooled data) from posterior inference (once per (date, horizon) pair, on single-date data) keeps the total computation tractable: pretraining dominates only in the number of gradient steps, while inference at each date takes roughly a millisecond on GPU.

An alternative would be to refit kernel hyperparameters independently for each (date, horizon) pair using only that date’s observations. A pilot experiment across three spatial random-90/10 splits finds no consistent improvement: mean per-well RMSE is 1.37 m for per-date refitting versus 1.33 m for pooled pretraining, and per-date NSE values are numerically unstable (range -37 to -92), reflecting ill-conditioned per-date fits with $n \approx 936$ observations but only 300 Adam steps of kernel optimization. The pooled procedure is therefore retained.

4.2.6. True-Observation Interpolation Baseline

The true-observation interpolation baseline replaces GRU predictions at training well locations with the true observed GWL values before fitting the GP. This provides a performance ceiling for the spatial interpolation stage: it answers the question of how well the GP could interpolate if the temporal forecasting stage were perfect.

The interpolation baseline gap measures how much the GRU’s forecasting errors propagate into the GP’s spatial predictions. If this gap is small, the bottleneck for further improvement lies in the GP’s spatial generalization ability, not in the GRU. If the gap is large, the temporal model should be improved first.

4.2.7. Jointly Trained Spatiotemporal Model

The jointly trained spatiotemporal model attempts to learn GRU weights that produce forecasts particularly suited for spatial interpolation. Rather than training GRU and GP independently, both components share a combined loss and are optimized simultaneously.

A three-way split of wells is required: training wells (for GRU and GP training), spatial validation wells (for GP supervision during training), and test wells. This differs structurally from the two-stage spatiotemporal pipeline, which uses only a two-way split into training and test wells. Because joint training reserves part of the training wells as a spatial validation set, the two configurations cannot be exactly matched on both splits. This structural asymmetry is an

inherent property of this approach and must be kept in mind when comparing results across configurations: matching the number of training wells gives the jointly trained spatiotemporal model fewer test wells for evaluation, while matching the test set reduces the GRU’s training data.

Early stopping uses patience = 15: the spatial validation loss descends non-monotonically for roughly 10–12 epochs before stabilizing, meaning the loss may temporarily increase over several consecutive epochs even while the overall trend is downward; patience = 5 would fire prematurely when the loss rises for a few consecutive epochs.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{GRU}} + \lambda \cdot \mathcal{L}_{\text{GP}} \quad (4.18)$$

where $\mathcal{L}_{\text{GRU}}$ is the MSE of GRU predictions at training wells, $\mathcal{L}_{\text{GP}}$ is the MSE of GP interpolations at spatial validation wells, and λ is a weight controlling the balance between the two losses. Crucially, the GP is conditioned on the GRU’s current predictions at training wells, so $\mathcal{L}_{\text{GP}}$ is an explicit function of the GRU’s parameters. This is possible because $\boldsymbol{\mu}_* = \mathbf{K}_*(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}\mathbf{y}$ is differentiable with respect to $\mathbf{y}$, so PyTorch’s autodiff traces through the Cholesky solve and returns gradients to the GRU without any special treatment. Within each epoch, two separate gradient passes update the GRU: Phase A backpropagates $\mathcal{L}_{\text{GRU}}$ directly from the temporal MSE on the training period; Phase B backpropagates $\lambda \cdot \mathcal{L}_{\text{GP}}$ through the GP’s conditioning equations back to the GRU, providing a spatial gradient signal. Phase B also updates the GP kernel hyperparameters; Phase A updates only the GRU.

The key structural difference from the two-stage spatiotemporal pipeline is that both the GRU temporal loss and the GP spatial loss are computed within each epoch, so gradients from both objectives update the GRU parameters.

The coupling weight $\lambda = 0.0001$ is selected by a systematic sweep over

$$\lambda \in \{0.00001, 0.0001, 0.001, 0.01, 0.1, 0.3, 0.5, 0.7, 0.9\}$$

on a single representative split seed; full results are reported in Table B.7. Performance improves monotonically as λ decreases from 0.9 to 0.0001, and degrades rapidly above 0.5. Below 0.0001 performance starts dropping again, so 0.0001 is the practical lower bound. The production joint training runs use the h256/14 architecture identified by manual grid search (the same as the two-stage spatiotemporal pipeline GRU), rather than h128/15 identified by automated HPO; a subsequent architecture sweep confirms h256/14 outperforms the automated HPO optimum under production settings (Section 6.3); Table 4.1 lists all structural differences and shared properties between the two configurations.

Aspect	Two-Stage	Jointly Trained
<i>Differences</i>		
Well split	2-way: train / test	3-way: train / val / test
GRU gradient signal	$\mathcal{L}_{\text{GRU}}$ only	$\mathcal{L}_{\text{GRU}}$ and $\lambda \cdot \mathcal{L}_{\text{GP}}$
GP supervision	None during GRU training	MSE at spatial validation wells
GP conditioning inputs	All training wells	Spatial train wells only
Early stopping patience	5 epochs	15 epochs
Batch size	4096	per-date
<i>Shared properties</i>		
GRU architecture	h256 / 4 layers, dropout 0.2	
GP kernel	Matérn-3/2 with ARD length-scales	
Evaluation target	Test wells, never seen during training	
Input features	Same GRU and GP feature sets	
GP conditioning dates	All test-period dates	

Table 4.1.: Structural comparison of the two-stage spatiotemporal pipeline and the jointly trained spatiotemporal model

4.2.8. Pipeline Comparison

Table 4.1 summarises the key structural differences and shared properties of the two-stage spatiotemporal pipeline and the jointly trained spatiotemporal model.

4.2.9. Evaluation Design Justification

The 40-seed, 1-GRU-seed, 1-GP-seed evaluation design reflects a deliberate balance between statistical robustness and computational cost. Table 4.2 summarises the rationale for each component.

Component	# seeds	Rationale
Split seed	40	Std $\approx 9 \times$ GP/GRU std; $40 \times 1 \times 1$ minimises total SE among all factorizations; $\rho = 0.603$ distance–error Spearman gives $t = 4.66$, $p < 0.001$
GP seed	1	Kernel init. noise negligible ($\sigma \approx 0.048$ nRMSE _{pw}); the seed with median nRMSE _{pw} across the 10 seeds tested
GRU seed	1	Stable at this data scale; split seed variance dominates

Table 4.2.: Multi-seed evaluation design: component counts and justification

Table 4.3 quantifies the variance contributed by each seed type, measured as nRMSE_{pw} with all other settings fixed.

GP seed std (≈ 0.048 nRMSE_{pw}) is small relative to the effect sizes compared across pipelines; a single median seed (GP seed 7, $z = -0.05\sigma$ from the 10-seed mean on pooled RMSE) is therefore sufficient. GRU seed std (≈ 0.038 nRMSE_{pw}) is comparable in magnitude and substantially

Seed type	# seeds tested	Std	Range
GP seed	10	0.048	0.136
GRU seed	11	0.038	0.146
Split seed	10	0.430	1.324

Table 4.3.: nRMSE_{pw} by seed type (std and range; all other settings fixed)

smaller than split seed std, so a single well-tuned seed (GRU seed 47, $z = -0.30\sigma$ from the 10-seed mean) is used throughout. Split seed std ($\approx 0.430 \text{ nRMSE}_{\text{pw}}$) is roughly nine times the GP seed std and more than eleven times the GRU seed std, which confirms that test set composition is the dominant source of uncertainty. Willard and Varadharajan (2025) find the same pattern in stream temperature prediction: architecture and input-feature diversity contribute more to ensemble performance than varying random weight initialization. Forty seeds (ss42–ss81) reduce the standard error of the mean to $0.430/\sqrt{40} \approx 0.068 \text{ nRMSE}_{\text{pw}}$, which is 95% of the reduction attainable with 81 seeds at half the computational cost. Because split variance dominates so strongly, the $40 \times 1 \times 1$ allocation minimises total SE among all integer factorizations of 40 total runs: diverting runs to additional GP or GRU seeds would only increase overall uncertainty. The design also provides strong power for the distance–error analysis: the Spearman test gives $t = 4.66$ ($p < 0.001$, $df = 38$).

4.3. Implementation

The full implementation is available at <https://github.com/calgo-lab/gist>.

4.3.1. Software Stack

The implementation uses Python 3.12 throughout. The GRU temporal model is implemented in PyTorch (Paszke et al., 2019), using standard `nn.GRU` layers with a custom sequence-to-sequence wrapper and static-feature initialization of the encoder hidden state. The Gaussian Process is implemented using GPyTorch (Gardner et al., 2018) with exact inference via Cholesky factorisation, as described in Section 4.1.2.

Hyperparameter optimization is managed via iterative grid search, as described in Section 4.1.4. Data preprocessing, feature engineering, and experiment tracking are handled with NumPy, Pandas, and a lightweight custom experiment registry. Visualisation uses Matplotlib and Seaborn.

4.3.2. Compute Infrastructure

All training and evaluation experiments are run on NVIDIA A100, H100, and H200 GPUs provided through the Data Science Kubernetes cluster at Berliner Hochschule für Technik (BHT).

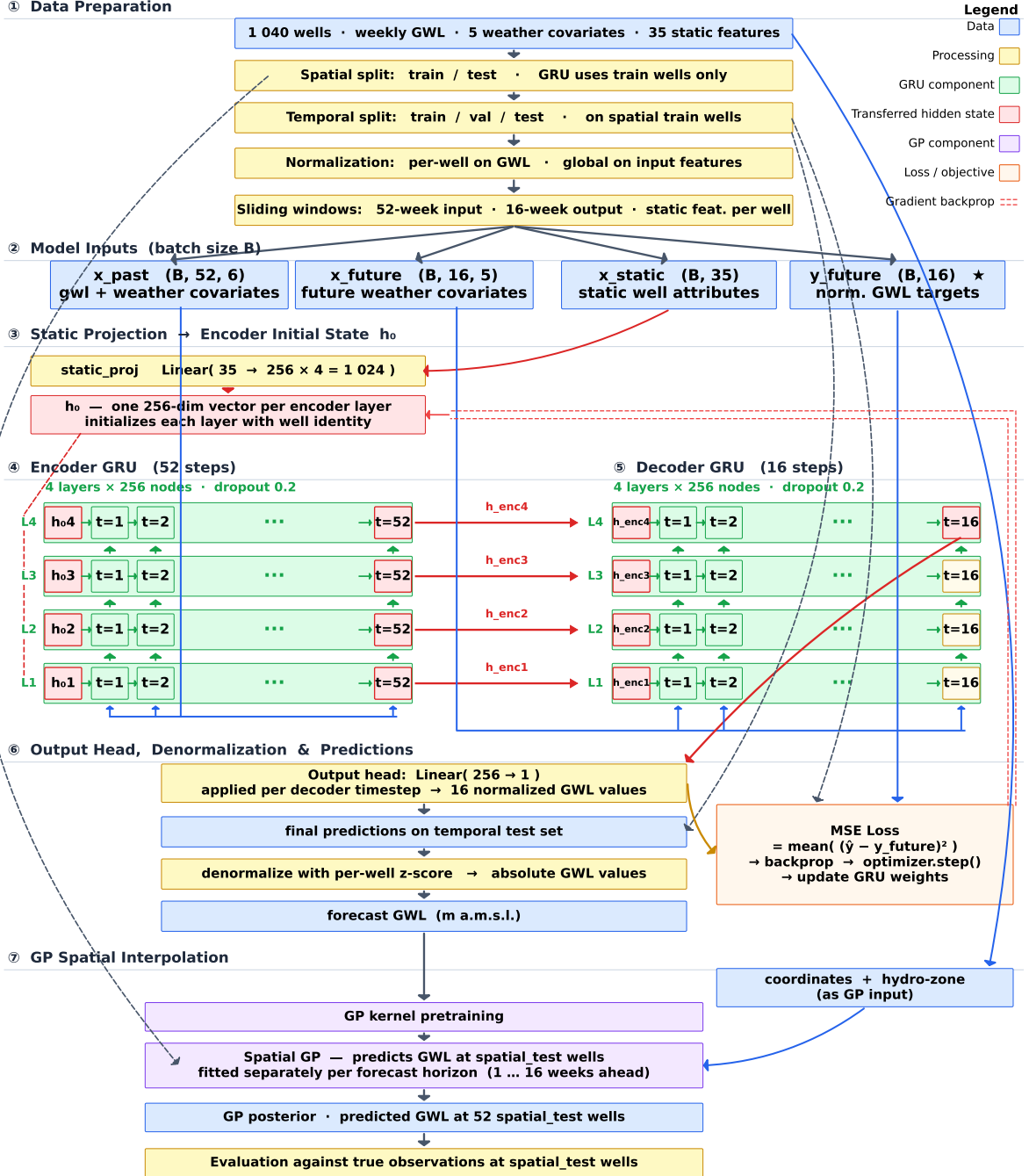


Figure 4.3.: Full two-stage spatiotemporal pipeline architecture (sections ①–⑦). The GRU forecasts GWL at training wells (sections ①–⑥); the GP then spatially interpolates those forecasts to test wells (section ⑦). Both components are trained independently.

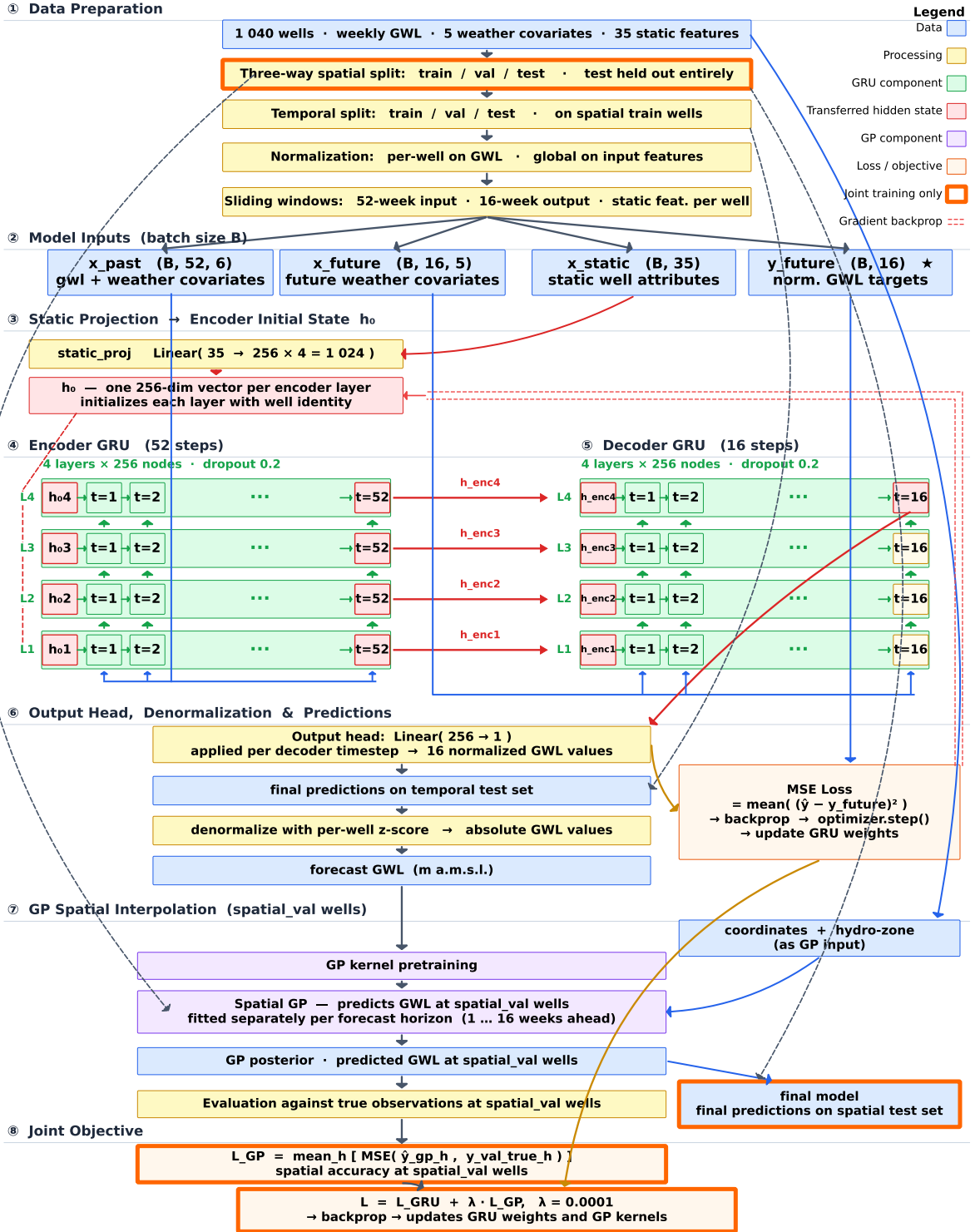


Figure 4.4.: Jointly trained spatiotemporal model (sections ①–⑧). GRU and GP are trained end-to-end with a combined loss $\mathcal{L} = \mathcal{L}_{\text{GRU}} + \lambda \cdot \mathcal{L}_{\text{GP}}$ ($\lambda = 0.0001$), combining GRU temporal MSE with GP spatial MSE at validation wells. The GP is conditioned on GRU predictions, so backpropagating $\mathcal{L}_{\text{GP}}$ returns gradients to the GRU. A three-way well split is required: GRU training wells, GP spatial validation wells, and test wells.

Each job runs inside a Docker container built from a custom image that pins the exact Python, PyTorch, and CUDA versions, ensuring that all experiments are fully reproducible regardless of which GPU node the cluster schedules the job on. Jobs are submitted via YAML-configured workloads specifying the container image, resource requests, and entry point.

The GRU training runtime is roughly 20 minutes per job (50 epochs with early stopping, 1,040 wells, batch size 4096). GP kernel fitting (300 gradient-descent steps on the marginal log-likelihood) is negligible in runtime; GP posterior evaluation (exact Cholesky inference across all forecast dates and horizons) takes approximately 3 minutes per split-seed. A full two-stage spatiotemporal pipeline job (1 GRU train + 1 GP evaluation) therefore takes approximately 24 minutes; the true-observation interpolation baseline approximately 3 minutes. The jointly trained spatiotemporal model, which runs GRU and GP end-to-end in a single job, takes approximately 51 minutes per run.

Table 4.4 summarises the per-pipeline cluster job structure and approximate runtimes per cluster job.

Configuration	Steps per cluster job	Approx. time/job
Two-Stage Spatiotemporal Pipeline	1 GRU train + 1 GP eval (gp_seed 7)	≈24 min
True-Observation Interp. Baseline	1 obs build + 1 GP eval	≈3 min
Jointly Trained Spatiotemporal Model	1 joint training GRU+GP run	≈51 min
Global GRU	1 GRU train (1,040 wells, no test set)	≈21 min

Table 4.4.: Cluster job structure and runtime for each model configuration

4.4. Chapter Summary

The GRU seq2seq model, Matérn-3/2 GP, and three pipeline variants are now fully specified. The evaluation protocol is fixed: a random 90/10 spatial split with a co-location constraint, repeated over 40 seeds, with hydrogeological zone as the sole GP feature (Table 4.1). Whether the two-stage pipeline actually benefits from the temporal forecasting step, and whether joint training helps at all, are open questions that Chapter 5 addresses empirically.

5. Results

5.1. Experimental Setup

All experiments use the configuration described in Chapter 4: the random 90/10 spatial split with 40 independent seeds, the GRU (h256/l4/do0.2), and the GP with Matérn-3/2 kernel and hydrogeological zone as the sole static feature. The primary reported metric is median per-well RMSE at $h = 16$; IQR-normalized nRMSE_{pw} is the primary metric for the GP feature ablation. All experiments use HYRAS 5.0 reanalysis data for meteorological inputs, including the 16-week forecast horizon; reported error values therefore represent an upper bound on achievable performance under perfect meteorological information (see Section 6.3.2).

5.2. GRU Temporal Performance

5.2.1. GRU vs. TFT: Replication Result

To establish a TFT baseline, the training procedure from code shared directly by the authors of Kunz et al. (2025) is replicated on the Brandenburg dataset. This replication differs from the original code in two ways. First, the original Kunz et al. code applies `fit_transform` on the validation set’s static feature covariates; this replication corrects that by applying `transform`, which uses the scaler parameters estimated on the training set rather than refitting on the validation population. Second, the original trains with distributed data parallelism across four GPUs, for an effective batch size of $4096 \times 4 = 16,384$; this replication uses a single GPU with batch size 4096. Validation-loss early stopping (patience = 5) is also added, which is not used in the original.

With these changes, the replicated TFT achieves an $h = 16$ RMSE of 0.089 m, compared to 0.093 m from the original code the authors of Kunz et al. (2025) shared for the Brandenburg dataset, a 4.3% improvement.

Table 5.1 compares both models on temporal forecasting at 1,040 monitored training wells during the test period, using the same evaluation approach the authors of Kunz et al. (2025) apply to the Brandenburg dataset, with 10 seeds each. All values are the median per well and then the mean across seeds; NSE uses the per-well test-period mean as baseline (Section 4.1.3). The GRU reaches $h = 16$ RMSE of 0.088 m and $h = 16$ NSE of 0.764, matching the replicated TFT to within 0.001 m, less than half the GRU’s seed-to-seed standard deviation ($\text{std} \approx 0.002$ m across

10 GRU seeds). KGE at $h = 16$ is 0.803 for the GRU and 0.814 for the TFT (Table 5.1), which suggests the small RMSE gap does not reflect systematic bias or variability differences. The roughly $90\times$ training speedup (from about 31 hours to about 21 minutes on an A100 GPU on the BHT Kubernetes cluster) is what makes the full experimental scope of this thesis tractable: the HPO runs, feature ablations, multi-seed analysis, and joint training experiments would all have been too slow at TFT training cost.

Model	RMSE (m)		NSE		IQR-nRMSE		KGE		vs. Kunz TFT	
	h=16	avg	h=16	avg	h=16	avg	h=16	avg	Δ RMSE	Δ NSE
TFT (Kunz et al., 2025)	0.093	0.068	0.740	0.844	0.378	0.284	0.814	0.881	—	—
TFT (replicated)	0.089	0.065	0.763	0.852	0.358	0.273	0.824	0.880	−4.3%	+3.1%
Global GRU (h256/l4/do0.2)	0.088	0.066	0.764	0.852	0.371	0.284	0.803	0.869	−5.4%	+3.2%

Table 5.1.: GRU vs. TFT: temporal performance (avg = mean over $h=1-16$; Δ = % change in $h=16$ metric relative to Kunz TFT)

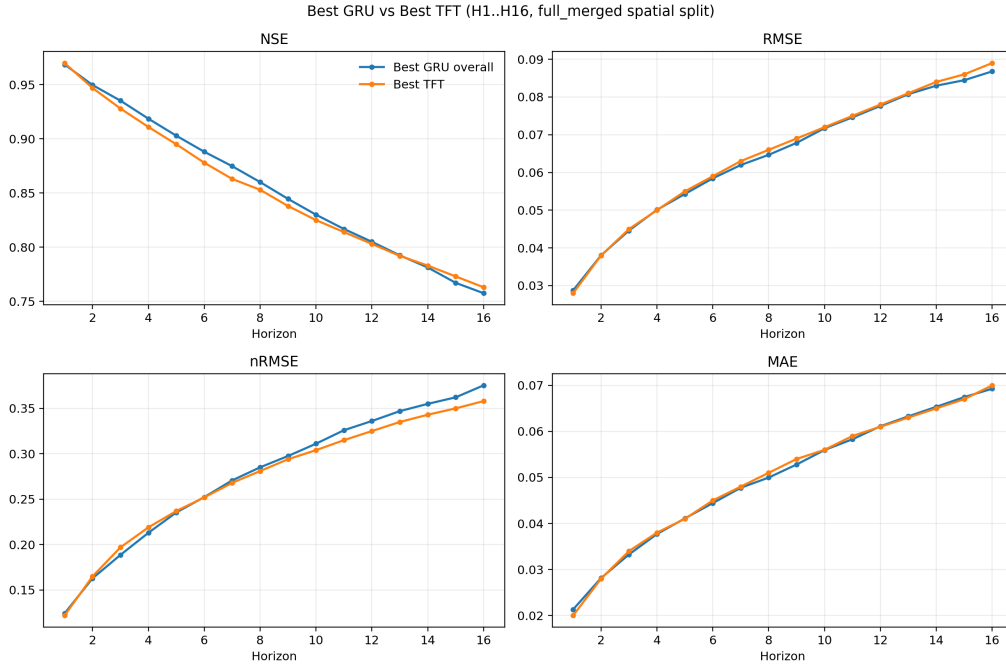


Figure 5.1.: Per-metric comparison of best GRU and best TFT across forecast horizons H1-H16. Both models show near-identical NSE, RMSE, nRMSE, and MAE trajectories.

Figure 5.2 shows the corresponding per-well RMSE and NSE at the training wells.

5.2.2. GRU Feature Permutation Importance

The Pardé seasonality coefficient dominates static feature importance (+47.3%); temperature and precipitation dominate dynamic feature importance (+91.5% and +84.5% respectively). Tables B.12 and B.13 in Appendix B.3 report the full rankings.

The dominant feature, the Pardé seasonality coefficient (+47.3%), captures the seasonal precipitation-response pattern for each well’s catchment, and its contribution exceeds that of the next four

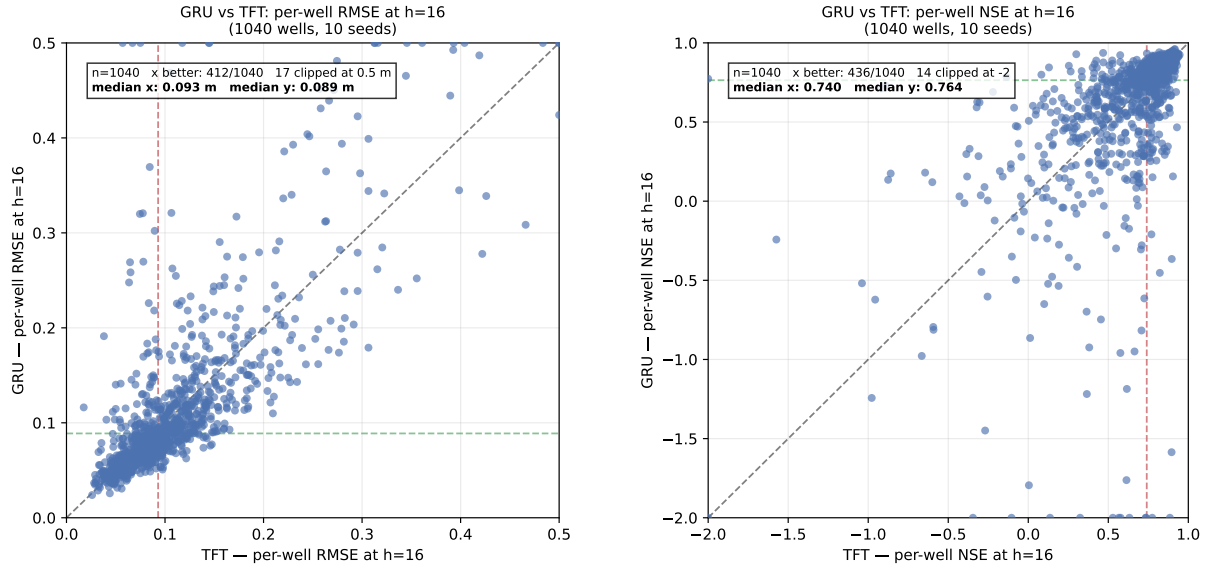


Figure 5.2.: Per-well RMSE (left) and NSE (right): GRU vs. TFT at training wells (10 seeds each)

highest-ranked features combined.

This dominance is specific to the h256/14 architecture. An earlier analysis on a shallower h128/15 model finds the Pardé seasonality coefficient to be harmful in that configuration, a discrepancy that likely reflects the h128/15 model lacking sufficient capacity to condition correctly on seasonal response patterns across all 1,040 wells. The h256/14 production runs retain the Pardé seasonality coefficient as an input; all 35 static features are included.

High permutation importance does not imply that a feature is essential. A GRU trained from scratch without the Pardé seasonality coefficient achieves essentially the same RMSE as one trained with it across 40 split seeds, within one seed-to-seed standard deviation. Permutation importance reflects how easy a feature is to exploit given the current model state, not how large its contribution to the final prediction could be if it were absent from training (Tables B.12 and B.13).

5.3. GP Feature Ablation

The GP feature ablation investigates the contribution of each static covariate to interpolation performance. The true-observation interpolation baseline is used throughout: a GP conditioned on true observed groundwater levels at training wells rather than GRU predictions. This keeps each feature’s contribution free of any influence from GRU forecast errors, and the results transfer directly to the two-stage spatiotemporal pipeline since both use the same GP kernel and feature encoding. The coordinate-only baseline (no static features) serves as the reference point.

Ablation protocol. All 30 feature configurations are evaluated on the random 90/10 split using 10 split seeds and 1 GP seed per configuration (300 GP evaluations in total). The configurations include hydrogeological zone, aquifer confinement, groundwater residence time, well screen depths, autocorrelation class, precipitation-response features, and groundwater level class, tested individually and in combinations. All numbers reported here are means over 10 split seeds. Full results for all 30 configurations are in Appendix B.1.

The ablation uses IQR-normalized nRMSE_{pw} as the primary metric. The 104 test wells cover a wide range of seasonal amplitudes (from $<0.5\text{ m}$ to $>10\text{ m}$); plain RMSE is dominated by high-amplitude wells, which carry more absolute error but respond differently to static features than low-amplitude wells. IQR normalization gives each well equal weight and produces a cleaner signal for feature selection. RMSE_{pw} is reported alongside for comparison.

Figure 5.3 summarises the relative performance of the core feature configurations; Table 5.2 shows the corresponding numerical values.

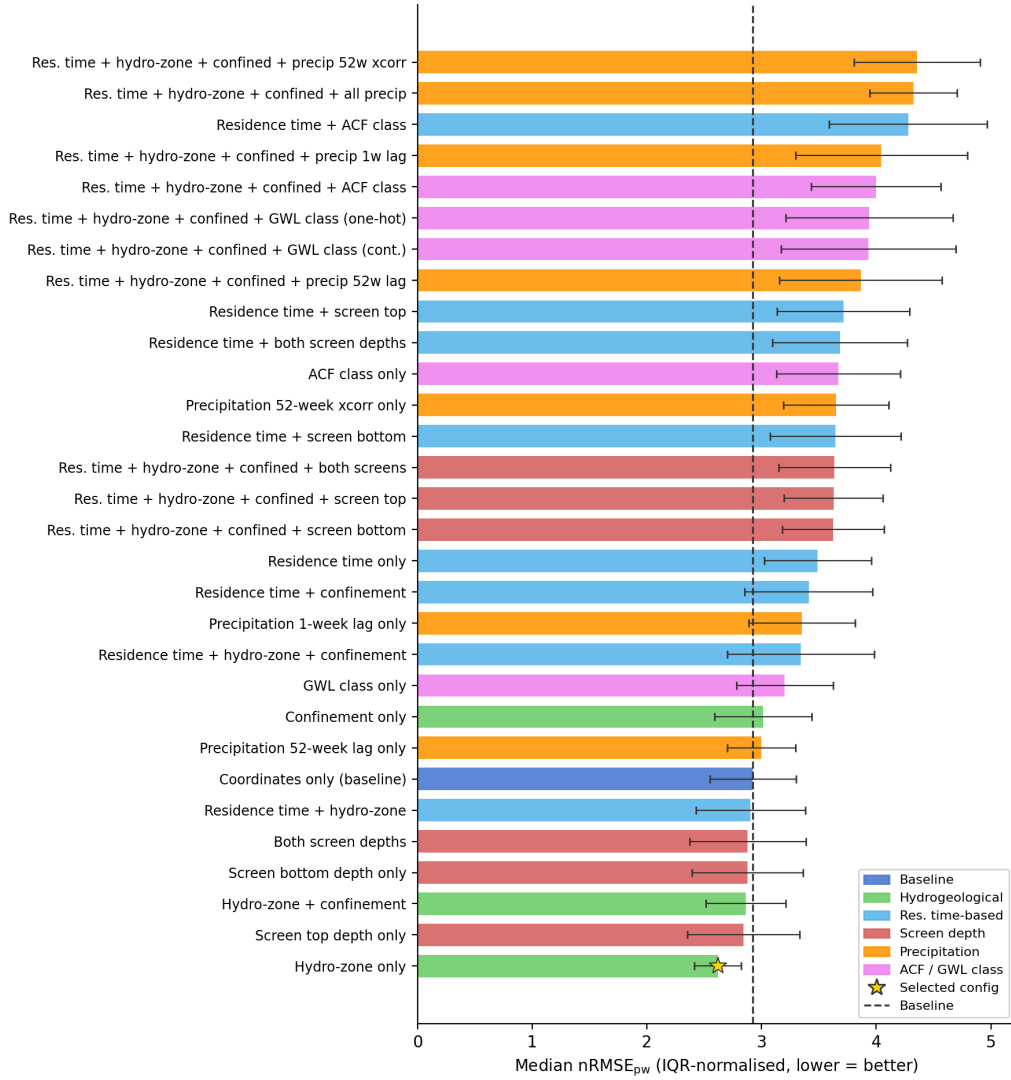


Figure 5.3.: GP feature ablation (IQR- nRMSE_{pw} , lower is better; dashed line: coordinate-only baseline; star: selected configuration)

Feature config	nRMSE	Δ nRMSE	RMSE (m)
Hydrogeological zone only *	4.444	−0.633	1.052
Screen top depth only	4.883	−0.194	1.070
Confinement only	5.017	−0.060	1.148
<i>Coordinates only (baseline)</i>	5.077	—	1.130
Hydrogeological zone + confinement	5.207	+0.130	1.153
Residence time only	5.437	+0.361	1.261

Table 5.2.: GP feature ablation: selected configurations (median IQR-nRMSE_{pw} and RMSE_{pw}; lower is better; full table in Appendix B.1)

Hydrogeological zone only is the selected configuration. The hydrogeological zone feature (one-hot encoded from three classes: recharge, transit, and discharge zones) achieves the largest reduction in IQR-normalized nRMSE_{pw}: -0.633 (-12.5%) relative to the coordinate-only baseline. No other single feature comes close: the next-best configurations show gains of only -0.060 to -0.194 on IQR-nRMSE_{pw}, all within one seed-to-seed standard deviation of the baseline, while the winning gap of -0.633 is more than three times the next-best gain. All precipitation-response features, residence time, and the autocorrelation class hurt performance. The selected configuration also achieves the best RMSE_{pw} simultaneously, winning on both metrics.

The nRMSE result is stronger than the corresponding RMSE comparison. The RMSE reduction is only -0.078 m (-6.9%), which is small relative to the seed-to-seed spread (std ≈ 0.16 m across 10 seeds). The difference between metrics is also visible in the rankings: confinement only ranks third on nRMSE ($\Delta = -0.060$) yet slightly worsens RMSE relative to the coordinate-only baseline ($+0.018$ m). Adding confinement to hydrogeological zone together is worse than hydrogeological zone alone on both metrics, with nRMSE increasing by $+0.130$ relative to the baseline.

Dynamic meteorological features. Temperature, precipitation, and relative humidity are tested on top of the best static configuration (hydrogeological zone only) using the true-observation interpolation baseline on the random 90/10 split, 10 split seeds and one GP seed per configuration (10 evaluations each). Table 5.3 reports the results. All three dynamic features increase median nRMSE_{pw} relative to the baseline (4.44): temperature by $+0.54$ ($+12\%$), relative humidity by $+0.98$ ($+22\%$), and precipitation by $+2.19$ ($+49\%$). Adding all three together does not improve over precipitation alone. The GP spatial kernel does not benefit from time-varying meteorological inputs, which is consistent with the kernel capturing spatial rather than temporal structure. Dynamic meteorological features are not added to the final GP configuration.

Matérn-5/2 kernel. The default kernel is Matérn-3/2. To check whether a smoother kernel would improve spatial interpolation, the true-observation interpolation baseline is re-run with Matérn-5/2 ($\nu = 2.5$) on the random 90/10 split (10 split seeds, 1 GP seed, hydrogeological zone only). Table 5.4 shows the comparison: Matérn-5/2 gives RMSE_{pw} of 1.235 ± 0.163 m vs.

Feature config	nRMSE (mean $\pm$ std)	Δ
<i>Hydrogeological zone only (baseline)</i>	4.44 ± 0.50	—
Hydrogeological zone + temperature	4.98 ± 1.06	+0.54
Hydrogeological zone + relative humidity	5.43 ± 1.12	+0.98
Hydrogeological zone + precipitation	6.64 ± 1.29	+2.19
Hydrogeological zone + all three met	6.63 ± 1.26	+2.19

Table 5.3.: GP dynamic feature ablation (median IQR-nRMSE_{pw} per well, mean over 10 split seeds; lower is better)

1.052 ± 0.160 m for Matérn-3/2. The smoother kernel is worse by 0.18 m on average (17.4%), which confirms Matérn-3/2 as the better choice.

Kernel	RMSE _{pw} (m)
Matérn-3/2 $\star$	1.052 ± 0.160
Matérn-5/2	1.235 ± 0.163

Table 5.4.: Matérn 3/2 vs. 5/2 kernel comparison for GP

5.4. Main Pipeline Results

5.4.1. Two-Stage Spatiotemporal Pipeline

The two-stage spatiotemporal pipeline is evaluated on the primary random 90/10 split using 40 split seeds (ss42–ss81), 1 GRU seed, 1 GP seed, the co-location-aware sampling algorithm, and hydrogeological zone only as the GP feature. The full metric comparison across all five model configurations is in Table 5.5; the discussion here focuses on the two-stage pipeline’s key results.

5.4.2. Two-Stage Pipeline and Interpolation Baseline

At $h = 16$, the two-stage pipeline RMSE of 1.082 m (Table 5.5) is roughly $12\times$ the GRU temporal RMSE of 0.088 m (Table 5.1).

The two-stage pipeline achieves a mean per-well RMSE of 1.119 ± 0.208 m and an IQR-normalized nRMSE_{pw} of 4.580 ± 0.909 (means $\pm$ std across 40 seeds). The true-observation interpolation baseline achieves 1.116 ± 0.208 m. The seed-to-seed standard deviation of the per-seed median gap is 0.033 m; dividing by $\sqrt{40}$ (for 40 split seeds) gives a standard error of 0.005 m. The observed mean gap of 0.003 m (less than 0.3%) is smaller than one standard error. A paired t -test across the 40 seeds gives $t_{39} = 0.68$, $p = 0.50$: if the true gap were zero, random sampling would produce a t -statistic at least this large 50% of the time, so the two configurations are effectively equal on this metric. Figure 5.4 shows the per-well RMSE comparison. The GP’s 95%

nominal prediction intervals achieve approximately 60% empirical coverage (mean 0.602 ± 0.043 across 40 seeds; PICP) at the spatial test wells (Section 6.3.2).

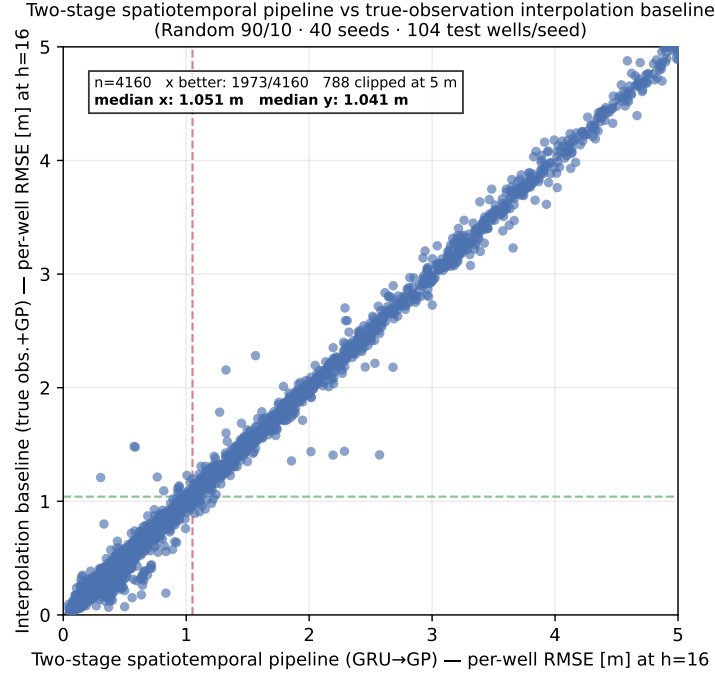


Figure 5.4.: Per-well RMSE: interpolation baseline vs. two-stage pipeline (random 90/10, 40 seeds; $n = 4,160$ (well, seed) pairs)

5.5. Spatiotemporal Pipeline Comparison

Table 5.5 compares four model configurations on the primary random 90/10 split (40 split seeds, co-location-aware sampling): one temporal baseline (Global GRU) and three spatiotemporal pipeline variants (1 GP seed, hydrogeological zone only).

GRU metrics are evaluated at monitored training wells during the test period and are not directly comparable to the GP pipeline metrics at 104 unseen test wells. The Interp. Baseline uses true observed GWL as GP input and gives an upper bound on pipeline performance.

Configuration	$h=16$ RMSE (m)	avg RMSE (m)	$h=16$ NSE	avg NSE
Global GRU (h256/l4/do0.2)	0.089 ± 0.002	0.069 ± 0.002	0.765 ± 0.011	0.849 ± 0.008
Two-Stage (GRU+GP)	1.082 ± 0.193	1.119 ± 0.208	-32.9 ± 16.4	-38.4 ± 16.9
Interp. Baseline	1.079 ± 0.191	1.116 ± 0.208	-33.1 ± 16.9	-38.6 ± 17.1
Jointly Trained	1.259 ± 0.290	1.286 ± 0.278	-46.1 ± 30.8	-52.1 ± 32.8

Table 5.5.: Pipeline comparison, random 90/10 split (40 seeds). Per-well NSE is expected to be strongly negative for all spatial pipeline variants; see Section 4.1.3 for the structural reason and Section 4.1.3 for the contrast with global NSE.

Figure 5.5 shows per-well RMSE comparing the global GRU temporal model against the two-stage pipeline.

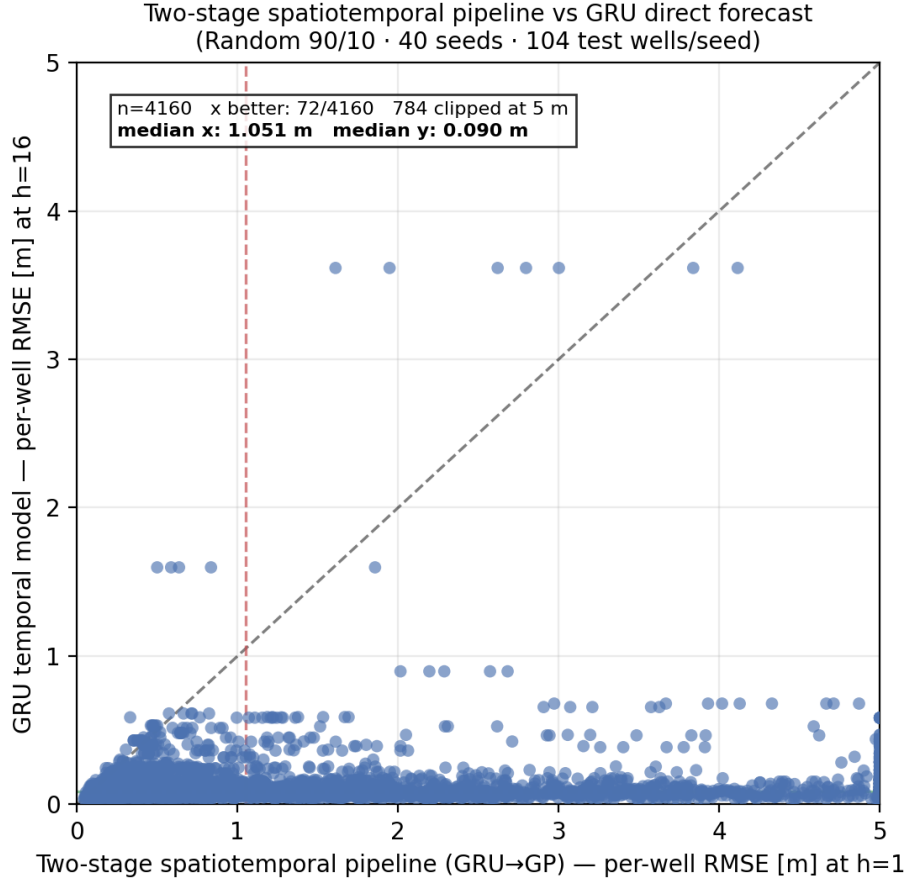


Figure 5.5.: Per-well RMSE: GRU temporal (y-axis) vs. two-stage pipeline (x-axis), random split (40 seeds)

Figure 5.6 shows median per-well RMSE across forecast horizons H1–H16 for all three pipeline configurations. All three show broadly flat profiles.

5.5.1. Joint Training

The jointly trained spatiotemporal model takes roughly twice as long to train as the two-stage pipeline and produces worse results on average: mean per-seed median $h = 16$ RMSE is 1.259 m vs. 1.082 m for the two-stage pipeline (16.4% higher).

Figure 5.7 shows per-well RMSE for all 4,160 (well, seed) pairs across the 40 seeds, clipped at 5 m. The two-stage pipeline is better for 2,199 of 4,160 pairs.

But the per-well breakdown tells a different story: the jointly trained model is worse for close wells and better for the most isolated ones (see Section 5.6).

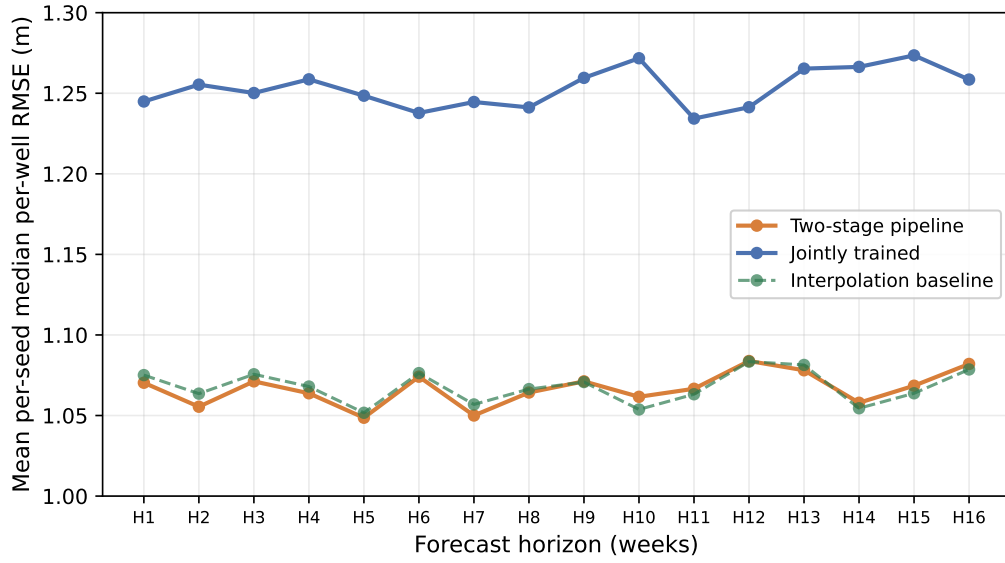


Figure 5.6.: Median per-well RMSE by forecast horizon (random 90/10, 40 seeds; all three pipeline configurations)

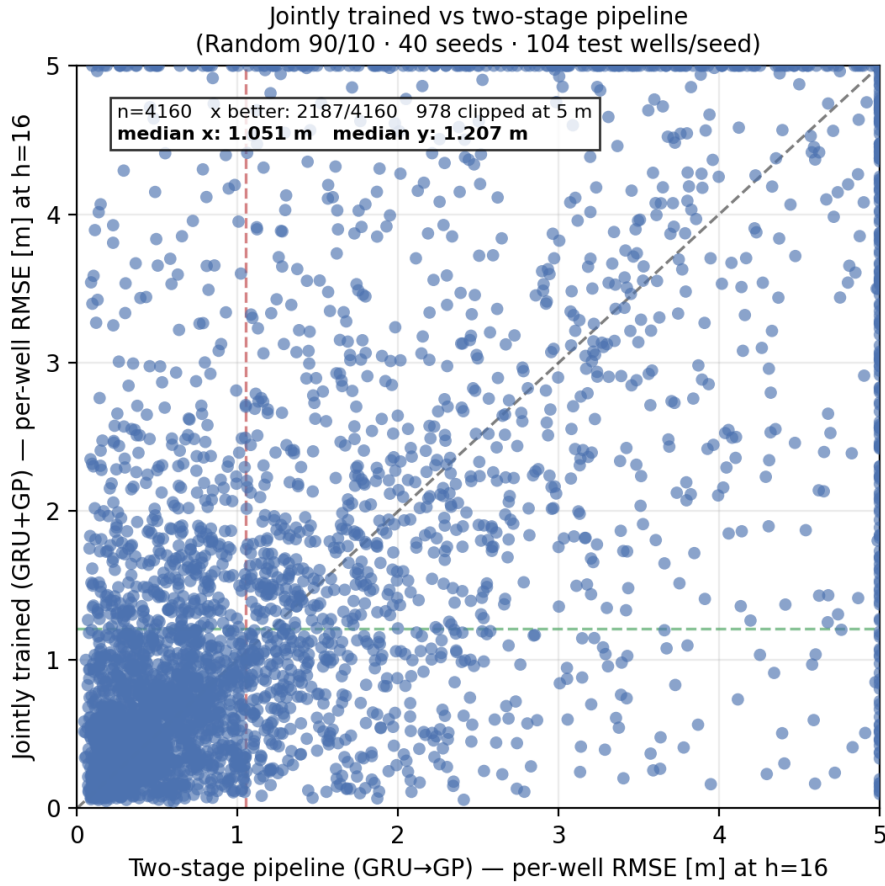


Figure 5.7.: Per-well RMSE: two-stage vs jointly trained (random 90/10, 40 seeds; $n = 4,160$ (well, seed) pairs; axis clipped at 5 m; two-stage better for 2,199/4,160 pairs)

5.6. Distance-Error Relationship

Distance between a test well and its nearest training neighbor is the primary driver of interpolation difficulty. This holds both across seeds and within seeds.

5.6.1. Across Seeds: Spearman Correlation

The random split exhibits substantial standard deviation across 40 seeds. The full per-seed breakdown of median nearest-training-well distance and RMSE_{pw} is in Table B.10 (Appendix B). Seeds that draw more spatially isolated test wells show higher RMSE: the Spearman correlation between median nearest-training-well distance and per-seed median RMSE is $\rho = 0.603$ ($p < 0.001$, $n = 40$, $t = 4.66$, $df = 38$). Figure 5.8 shows this relationship: each point is one of the 40 seeds, with its median nearest-training-well distance on the x -axis and its median two-stage pipeline RMSE on the y -axis.

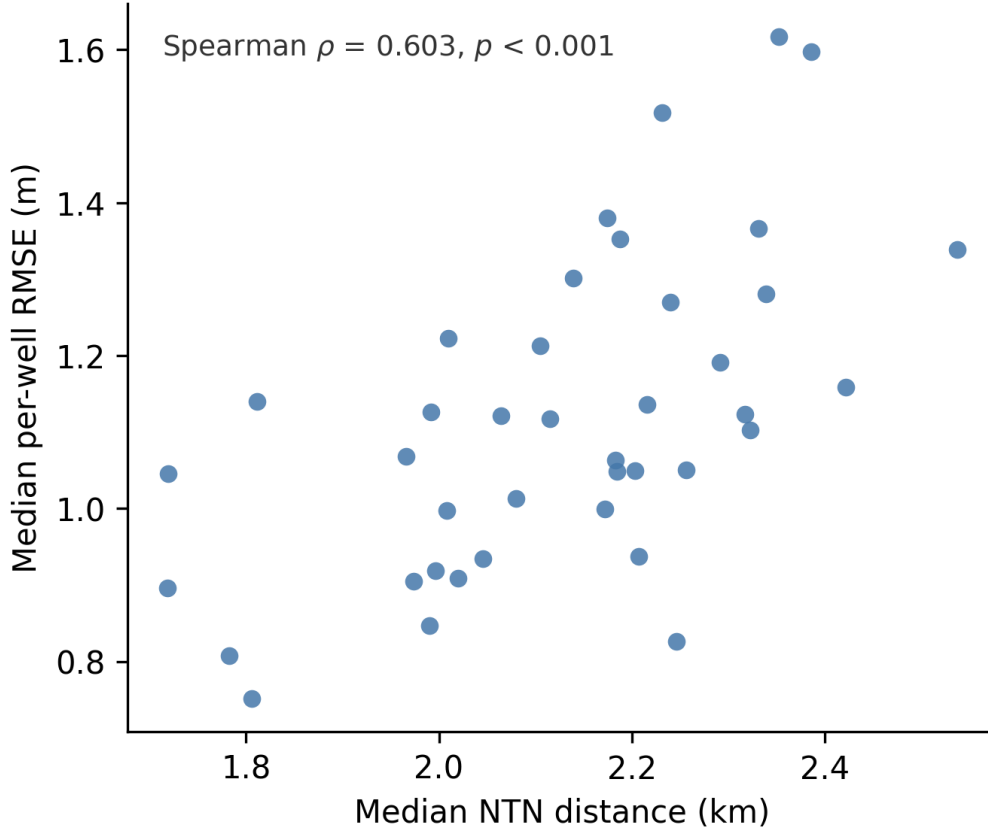


Figure 5.8.: Median RMSE vs. median nearest-training-well distance across 40 seeds (random 90/10 split; Spearman $\rho = 0.603$, $p < 0.001$)

5.6.2. Distance-Binned RMSE

Figure 5.9 shows median per-well RMSE in 1 km distance bins for both pipeline variants, pooled across 40 seeds. Below 2.5 km, the two-stage pipeline performs better. From roughly 4.5 km onward the ranking reverses: by the farthest distance bin (median distance 6.8 km) the two-stage pipeline reaches 5.4 m against 4.3 m for the jointly trained model. Figures 5.10a and 5.10b confirm this at the extremes: for the closest 10% of test wells (≤ 0.61 km, $n = 422$), the two-stage pipeline is better on 282 of 422 pairs (median RMSE 0.675 m vs. 1.098 m for the jointly trained model); for the farthest 10% (≥ 5.45 km, $n = 413$), the jointly trained model is better on 282 of 413 pairs (median RMSE 4.261 m vs. 5.369 m for the two-stage pipeline).

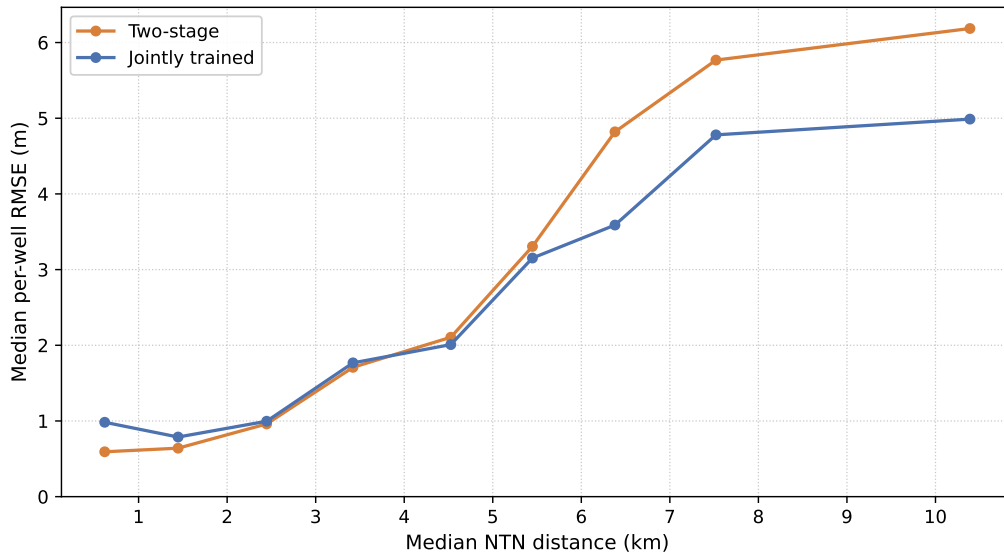


Figure 5.9.: Median per-well RMSE vs. nearest-training-well distance (km) for both pipeline variants (random 90/10, 40 seeds)

5.7. Alternative Split Strategies

The primary evaluation uses the random 90/10 split repeated over 40 seeds. The three alternative test set configurations described in Section 4.2.3 are evaluated on the same two-stage spatiotemporal pipeline to assess whether the choice of split design changes the substantive conclusions. All three use a 90/10 fraction and are run over 10 seeds.

Split strategy	Test wells	avg RMSE (m)	h=16 RMSE (m)	avg IQR-nRMSE
random 90/10 (primary, 40 seeds)	104	1.119 ± 0.208	1.082 ± 0.193	4.580 ± 0.909
Dense-area	104	0.678 ± 0.089	0.681 ± 0.109	2.900 ± 0.379
Sparse-area	104	5.102 ± 0.690	5.409 ± 0.803	22.269 ± 3.671
Cluster-stratified	98	1.215 ± 0.202	1.164 ± 0.220	5.123 ± 0.973
Random 80/20	208	1.240 ± 0.154	1.200 ± 0.142	5.084 ± 0.586

Table 5.6.: Two-stage pipeline under alternative split strategies (alternatives: 10 seeds each; primary: 40 seeds; lower is better)

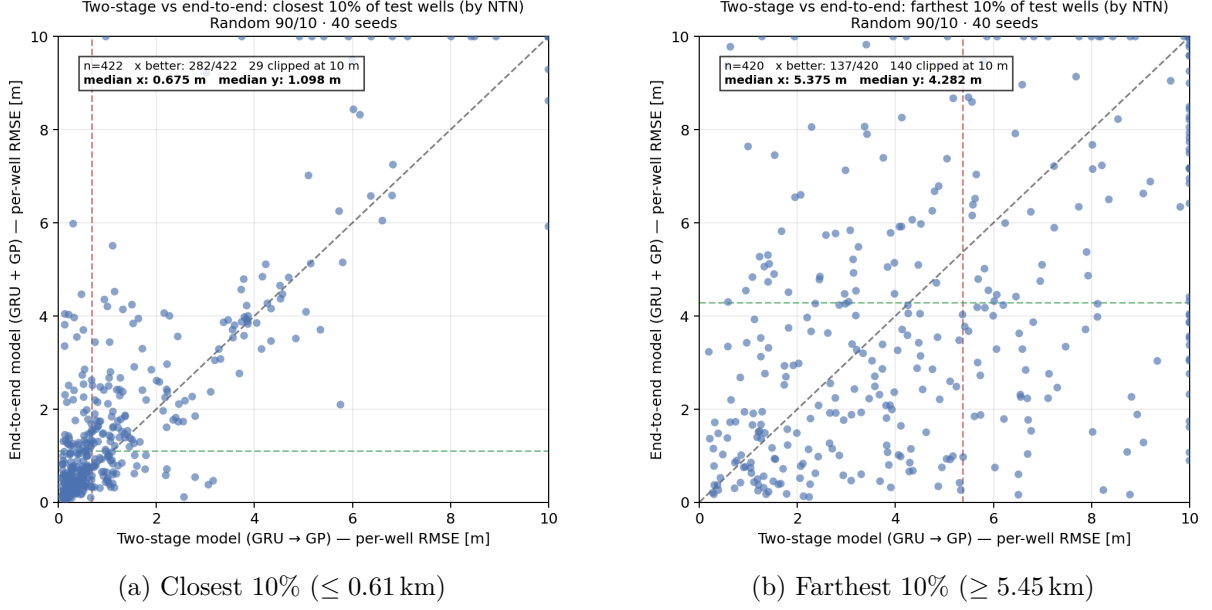


Figure 5.10.: Per-well RMSE: jointly trained vs. two-stage pipeline at the extremes of nearest-training-well distance (random 90/10, 40 seeds)

The Dense-area split achieves substantially lower RMSE than the random split at the same test fraction (0.678 m vs. 1.119 m). By restricting the eligible test pool to wells in the denser half of the monitoring network, the split makes it more likely that test wells have close training neighbors. The spatial interpolation task is therefore easier by construction.

The Sparse-area split restricts the eligible test pool to the 20% of wells with the largest nearest-training-well distances, which are the most spatially isolated wells in the Brandenburg monitoring network. The two-stage pipeline achieves avg RMSE of 5.102 ± 0.690 m on this split, roughly $4.6\times$ higher than on the random 90/10 (1.119 m). The jointly trained pipeline achieves 3.527 ± 0.406 m avg RMSE on the same split, 30.9% lower. The ordering from the primary evaluation is reversed, consistent with the distance-binned result in Section 5.5.1.

The Cluster-stratified split achieves avg RMSE of 1.215 ± 0.202 m on approximately 98 test wells, 8.6% higher than the primary random 90/10 split (1.119 m).

A random 80/20 split (208 test wells, 10 seeds) achieves avg RMSE of 1.240 ± 0.154 m, compared to 1.119 ± 0.208 m for the primary 90/10 split. The larger test fraction places more test wells in sparser parts of the monitoring network, raising error slightly. Full results for all split configurations are in Table B.2.

Figure 5.11 shows $h=16$ RMSE for the two-stage and jointly trained pipelines across all four split designs. For the random 90/10 split, the same 10 seeds (ss42–ss51) are used here as for the three alternative splits, rather than the full 40-seed production run, to keep the seed variance comparable across configurations.

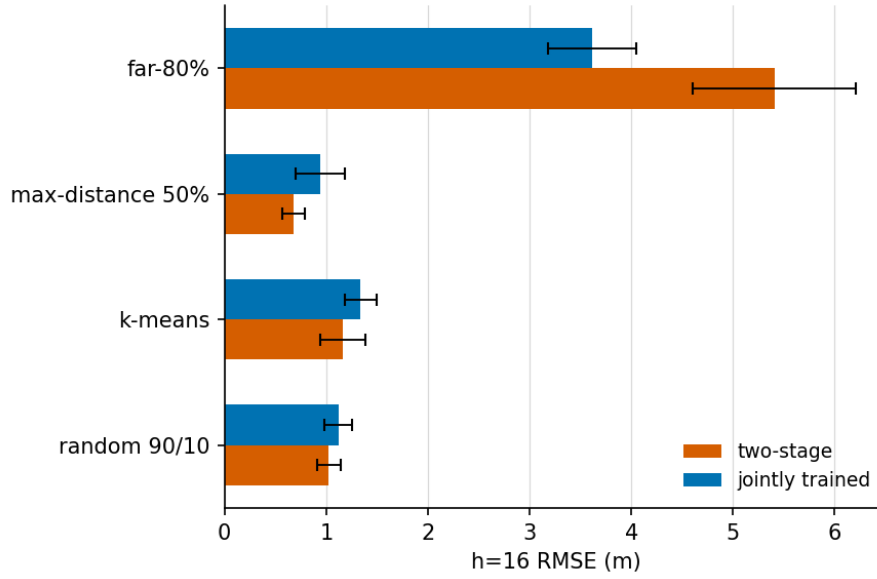


Figure 5.11.: $h=16$ RMSE for the two-stage and jointly trained pipelines across four split designs (10 seeds each; error bars show ± 1 standard deviation). The jointly trained spatiotemporal model wins only on the Sparse-area split, which selects the most spatially isolated test wells.

5.8. Interpolation Error by Hydrogeological Attributes

This section breaks down interpolation error by three static hydrogeological attributes: hydrogeological zone, aquifer confinement, and groundwater residence time. All analyses use the 1,040 (well, seed) pairs from the 10-seed random 90/10 split evaluation. A well appearing in multiple seeds' test sets contributes one pair per seed, because each draw produces a different training set, a different trained model, and therefore different predictions.

5.8.1. Hydrogeological Zone

Brandenburg's monitoring network is partitioned into three zone types: discharge zones (56% of pairs), transit zones (36%), and recharge zones (8%). Table 5.7 reports the median RMSE_{pw} and the median nearest-training-well distance for each zone.

Zone	Pairs	Median RMSE (m)	Nearest-training-well dist. (km)
Discharge zones	583	0.636	1.609
Transit zones	374	2.248	2.679
Recharge zones	83	7.972	3.387

Table 5.7.: Interpolation RMSE by hydrogeological zone

Discharge zones achieve the lowest median RMSE (0.636 m) and the shortest median nearest-training-well distance (1.609 km). To separate the zone effect from data density, Table 5.8 subsamples each group by trimming pairs with short nearest-training-well distances from the

bottom of each distribution until the median nearest-training-well distance is nearly equal across zones.

Zone	Pairs	Median RMSE (m)	Nearest-training-well dist. (km)
Discharge zones	180	1.115	3.389
Transit zones	270	2.320	3.410
Recharge zones	83	7.972	3.387

Table 5.8.: Distance-matched RMSE by hydrogeological zone

At matched median nearest-training-well distances, recharge zones are roughly $7\times$ harder to interpolate than discharge zones (7.972 m vs. 1.115 m median RMSE_{pw}). Hydrogeological zone thus affects interpolation difficulty beyond what spatial isolation explains.

5.8.2. Aquifer Confinement

Table 5.9 splits pairs by aquifer confinement type.

Confinement	Pairs	Median RMSE (m)	Nearest-training-well dist. (km)
Unconfined	769	1.056	2.106
Confined	271	1.005	2.113

Table 5.9.: Interpolation RMSE by confinement type

Median RMSE is nearly identical for unconfined (1.056 m) and confined (1.005 m) aquifers at essentially the same median nearest-training-well distance (2.106 km vs. 2.113 km). Aquifer confinement is not a meaningful predictor of interpolation difficulty.

5.8.3. Groundwater Residence Time

Table 5.10 bins the continuous residence time variable into four intervals. Of the 1,040 pairs, 43 have missing residence time values and are excluded.

Residence time	Pairs	Median RMSE (m)	Nearest-training-well dist. (km)
0–5 yr	299	0.759	1.946
5–15 yr	417	1.057	2.171
15–30 yr	246	1.096	2.121
30–50 yr	51	2.129	1.741

Table 5.10.: Interpolation RMSE by residence time

Median RMSE increases monotonically with residence time, from 0.759 m (0–5 years) to 2.129 m (30–50 years). Notably, the 30–50-year bin has the shortest median nearest-training-well distance (1.741 km) yet the highest median RMSE. Table 5.11 subsamples each group to a common median nearest-training-well distance of 2.10 km (the overall median across all pairs) to control for this.

Residence time	Pairs	Median RMSE (m)	Nearest-training-well dist. (km)
0–5 yr	283	0.766	2.104
5–15 yr	401	0.970	2.101
15–30 yr	246	1.096	2.121
30–50 yr	41	3.425	2.113

Table 5.11.: Distance-matched RMSE by residence time

After distance matching the gradient strengthens: the 30–50-year bin reaches 3.425 m, versus 0.766 m for the 0–5-year bin ($4.5\times$ higher). The residence-time effect is therefore not explained by data density.

5.9. Chapter Summary

The results point consistently in one direction: temporal forecast quality is not what limits the spatiotemporal pipeline. Spatial generalization is, and it degrades with distance from the training set. Hydrogeological zone is the only static feature that reliably helps the GP. Joint training does not solve the spatial problem on average but gains an advantage at the extreme tail of spatial isolation. What these findings mean for the forecast-anywhere goal, which approaches failed and why, and where the main gaps are, is the subject of Chapter 6.

6. Discussion

Section 6.1 identifies two findings that run through the results: spatial isolation as the primary driver of interpolation difficulty, and the GP as the primary bottleneck. Section 6.2 draws evaluation design lessons, and Section 6.3 documents approaches that did not improve results along with the limitations of this work.

6.1. Spatial Isolation and the GP Bottleneck

The first finding is that spatial isolation drives interpolation difficulty: seeds and splits that draw spatially isolated test wells produce systematically higher error, and the distance–error relationship holds both across seeds and within seeds, meaning that within any given seed, wells farther from training neighbors also show higher error (Section 5.6). The second is the GP bottleneck: the GRU’s temporal predictions at training wells are already accurate enough that substituting them with true observations changes interpolation error by less than 0.003 m (Section 5.4.2), so progress on the forecast-anywhere problem requires better spatial modeling, not a better temporal model.

6.1.1. Spatial Isolation Drives Interpolation Difficulty

Recharge zones are characterized by variable infiltration rates and subsurface properties that produce GWL dynamics that change more sharply over short distances than in discharge zones, where groundwater flows more uniformly, so nearby wells tend to share similar levels. The GP’s Matérn-3/2 kernel uses a single smoothness parameter for the whole study area, which means it treats spatial similarity as decaying at the same rate everywhere; recharge zones are where GWL can change sharply over short distances, a pattern a single smoothness parameter cannot fit. Hydrogeological zone works as a GP feature because it groups wells that tend to have similar GWL dynamics, beyond what their location alone would predict. All other tested feature groups either give smaller improvements or even degrade performance relative to the coordinate-only baseline.

6.1.2. The GP Bottleneck

Running 40 split seeds across hundreds of GP configurations would not have been practical with TFT at its training cost; the GRU’s speed makes that evaluation affordable. The multi-seed protocol that establishes the distance–error relationship and quantifies seed-to-seed variance is only possible because the temporal backbone is fast enough to run many times. The speedup’s main contribution is to evaluation methodology, not to spatial interpolation quality.

Lim et al. (2021) show in their ablation that self-attention is among the most important components of TFT. The GRU has no self-attention, yet matches TFT on Brandenburg. This suggests the dataset’s smooth seasonal dynamics do not require attention over long time spans.

6.2. Evaluation Design Lessons

6.2.1. The Co-Location Constraint

Without the co-location constraint (forcing wells within 8 m of each other into the training set), the spatiotemporal two-stage pipeline can achieve misleadingly low RMSE on some test wells by effectively copying a co-located training well. The reported metric is dominated by trivially easy cases, not by genuine spatial interpolation. Any spatial evaluation protocol that uses random well splitting without checking for co-located wells may systematically overestimate interpolation performance.

6.2.2. Multi-Seed Evaluation and Spatial Composition

The random split exhibits substantial standard deviation across seeds. The primary cause is that seeds differ in the median distance from test wells to their nearest training neighbor: seeds that draw spatially isolated test wells face a fundamentally harder interpolation problem. This is confirmed by the seed-level analysis in Section 5.6, which shows a positive Spearman correlation ($\rho = 0.603$; Section 5.6) between median nearest-training-well distance and per-seed RMSE. A study relying on a single split seed may therefore report metrics that reflect its spatial composition as much as model quality.

6.2.3. Alternative Splits Confirm, Not Replace, the Post-Hoc Distance Analysis

The sensitivity analysis in Section 5.7 shows that the Dense-area, Sparse-area, and Cluster-stratified splits all reach qualitatively consistent conclusions with the primary random split. Because each alternative split draws a different training set, the results come from independently trained models rather than post-hoc reselections of the random split, which makes the agreement a genuine robustness check. The post-hoc seed analysis on the random split is the more general

characterisation, since it spans the full range of spatial configurations rather than fixing difficulty to a chosen range. None of the three alternatives is appropriate as a primary evaluation: the Dense-area split has artificially low RMSE, the Sparse-area split has artificially high RMSE, and the Cluster-stratified split’s test-well count is not exactly controllable since cluster sizes are discrete.

6.3. Limitations and Dead Ends

6.3.1. Negative Results and Dead Ends

The following directions do not improve results. They are documented because negative results narrow the hypothesis space and prevent revisiting disproved approaches.

Residual GP ($nRMSE \approx 192$)

Training the GP on residuals (the difference between true observations and GRU predictions at training wells) rather than raw GWL values fails completely: $nRMSE_{pw} \approx 192$.

GRU prediction errors are not spatially correlated. They are well-specific, as each well has its own dynamics that the global model cannot fully capture (e.g., local recharge anomalies, well-specific seasonal timing). A residual GP therefore fits noise rather than a spatially coherent field and extrapolates that noise catastrophically to test wells.

There is a positive reading of this failure. The GRU’s predictions at each well are accurate enough that the residuals show no spatial pattern. If the GRU leaves a systematic trend across locations, a GP could pick that up; since the errors are well-specific rather than regional, the residual GP is just fitting noise.

Surface Elevation Clipping

A hypothesis is investigated that clipping GWL predictions to ground surface elevation (GSE, i.e. $\hat{y} \leftarrow \min(\hat{y}, GSE)$) might improve performance, since physically the water table cannot rise above the ground surface. Testing GSE clipping on all evaluated model configurations yields $\Delta = 0.000$ in all cases: the model never predicts GWL above the ground surface, so the clipping constraint is never active.

Depth to Groundwater Table as GP Feature

Following a suggestion by a collaborator at BGR, depth to the groundwater table (DGT) is tested as an additional GP spatial feature on the random split in a dedicated side experiment; because this experiment is conducted after the main GP feature ablation protocol has been finalized, the result is reported here rather than in Section 5.3. DGT alone is harmful ($\text{nRMSE}_{\text{pw}} \approx 8.66$ vs. 4.444 for hydrogeological zone alone); combined with hydrogeological zone it also performs worse than hydrogeological zone alone ($\text{nRMSE}_{\text{pw}} \approx 6.98$ vs. 4.444). DGT does not improve spatial interpolation and is excluded from the selected feature set.

6.3.2. Limitations

Hyperparameter search scope. Hyperparameters for the GRU and GP are found via iterative grid search. For the jointly trained model, automated HPO with Optuna (Akiba et al., 2019) finds h128/15 with coupling weight $\lambda = 0.034$, but a manual architecture sweep shows h256/14 with $\lambda = 10^{-4}$ gives better interpolation performance. The automated result is unreliable for two reasons: Optuna ran only a limited number of trials, and each trial was evaluated on a single spatial split, where variance is high enough to flip architecture rankings (Section 4.2.9).

HYRAS reanalysis weather data. All experiments use HYRAS 5.0 reanalysis data for the meteorological forcing inputs, including the 16-week forecast horizon. HYRAS 5.0 is a retrospective reconstruction that is available only for the past; it is not a weather forecast. In operational deployment, accurate precipitation and temperature forecasts over 16 weeks are unavailable, and forecast uncertainty would propagate into GWL predictions. The RMSE figures reported here therefore represent an upper bound on achievable performance under perfect meteorological information. The gap between HYRAS-driven and forecast-weather-driven performance is not quantified in this thesis.

GP calibration failure. The GP’s 95% nominal prediction intervals achieve only approximately 60% empirical coverage (mean 0.602 ± 0.043 across 40 seeds; Prediction Interval Coverage Probability, PICP) at spatial test wells (production two-stage spatiotemporal pipeline). This is a structural property of GP posterior inference applied to spatial extrapolation: kernel hyperparameters are optimized on training-well data, which may have a different spatial density and correlation structure than the test region. Overconfident here means the kernel-derived posterior intervals are too narrow: the posterior variance is bounded by the output scale σ_f^2 learned on training data and does not automatically widen for isolated test locations where the actual prediction error exceeds that prior scale. Note that poor coverage is not a direct consequence of poor point predictions: a model with the same NSE but wider uncertainty intervals would still achieve 95% PICP. The low coverage reflects that the GP underestimates its own uncertainty, which is a separate issue from prediction accuracy.

Static feature availability at genuinely unmonitored locations. The GP relies on a single static hydrogeological feature, hydrogeological zone, selected by the feature ablation in Section 5.3. Residence time and aquifer confinement are evaluated in the ablation but excluded because both degrade interpolation performance relative to the coordinate-only baseline. At genuinely unmonitored locations (where no well has ever been installed), hydrogeological zone may not be available or may require spatial interpolation from the nearest monitoring records. The accuracy of such interpolated features, and their effect on GP predictions, is not evaluated in this thesis.

Single study area. All experiments are conducted exclusively on the Brandenburg LfU monitoring network. The spatial density of this network (1,040 wells over 29,654 km²), the hydrogeological diversity of the region, and the HYRAS 5.0 meteorological forcing are specific to this setting. Whether the GRU-GP pipeline and the optimal feature set generalize to other regions, countries, or monitoring densities remains an open question. The pipeline architecture is likely to transfer to other settings, but the feature selection is less certain. Hydrogeological zone works here because it proxies for the spatial correlation structure of GWL dynamics in Brandenburg. In regions where no analogous categorical variable captures that structure, or where GWL dynamics do not follow the same zone pattern, the GP is reduced to coordinate-only interpolation and performance would likely be closer to the coordinate-only baseline in Table 5.2.

GRU seed variance not fully characterized. The multi-seed evaluation uses a single GRU seed (the best HPO configuration, on the median performing seed out of 10 GRU model seeds) for all spatial interpolation experiments. While GRU performance is stable at this data scale (dominated by split seed variance rather than model initialization), the full sensitivity of the spatial interpolation results to GRU seed choice is not quantified. The same applies to the GP: all spatial experiments use a single GP seed, and sensitivity to GP random initialization is not quantified.

Joint model feature parity. The original 40-seed joint runs use 32 GRU static inputs, as hydroraum one-hot features are present in the GP but absent from the GRU, while the two-stage pipeline uses 35. Additional re-runs with matching inputs ($n_{\text{static}} = 35$) show no meaningful difference (mean $\Delta\text{RMSE}_{\text{pw}} = -0.011$ m, paired t -test $p = 0.311$), so the original joint results are valid for comparison.

Joint training per-date batching. Both Phase A and Phase B of joint training iterate over dates, taking all training wells active on a given date as a batch. Per-date batching is necessary for Phase B because the GP conditions on all training-well predictions at a single point in time. It is not necessary for Phase A, which is standard GRU temporal training with no spatial component and could instead use random batching at a larger batch size, as the two-stage GRU does. This would reduce the number of gradient steps per epoch in Phase A substantially and speed up joint training.

Metric sensitivity. All feature comparisons in Section 5.3 use median per-well nRMSE as the primary criterion because IQR normalization accounts for wells with very different absolute GWL variance. Using per-well RMSE instead could favour different features: a feature that concentrates gains at high-variance wells looks better under nRMSE than under absolute RMSE. This sensitivity between the two metrics is not systematically explored across all feature combinations.

Co-location constraint on alternative splits. The alternative split strategies in Section 5.7 are also affected by the co-location constraint. Around 30% of wells are pre-assigned to the training set before any split procedure runs, since co-located pairs always go to training. The pool of wells that the Dense-area, Sparse-area, and Cluster-stratified splits actually operate on is therefore already filtered. Co-located wells make up around 15% of unique physical locations, so the effect is limited, but all three splits are working from an already-filtered pool, not the full network, which means none of the three split strategies operates on the complete well set as designed.

Limitations of the random split. The primary random split is not fully random: the co-location constraint pre-assigns a subset of wells to the training set before random selection runs on the remainder. This is unavoidable unless all but one well per group of co-located wells is dropped, and the effect on aggregate RMSE is small since co-located wells account for a small fraction of unique locations. The reported results should be understood as coming from a random draw over the subset of wells not pre-assigned by the co-location constraint, not the full network.

6.4. Chapter Summary

This chapter’s two main conclusions are that spatial isolation drives interpolation difficulty and that the GP, not the GRU, is where most of the pipeline’s error comes from. Recharge zones are roughly 7× harder to interpolate than discharge zones at matched distances (Section 5.8.1). Three evaluation lessons follow: the co-location constraint prevents trivially easy test cases, multi-seed evaluation is necessary because spatial composition dominates variance, and metric choice can reverse feature rankings. None of the dead-end approaches documented here improve on the standard two-stage pipeline, and feature selection is likely less transferable than the pipeline architecture since hydrogeological zone works here because it proxies a spatial correlation structure specific to Brandenburg. The following chapter draws together the main findings and outlines future directions.

7. Conclusion

7.1. Summary of Findings

This thesis presents GIST (Groundwater Interpolation in Space and Time), a spatiotemporal pipeline for groundwater level forecast interpolation across Brandenburg. The core questions are what limits the pipeline’s accuracy and what determines how hard any given prediction is.

A GRU sequence-to-sequence model matches the Temporal Fusion Transformer of Kunz et al. (2025) on the Brandenburg dataset, with a median per-well RMSE of 0.088 m at $h = 16$ against 0.093 m for the TFT, at roughly $90\times$ lower training cost. That speedup makes everything else in this thesis feasible (Table 4.4). The hyperparameter optimization, the feature ablation studies, the multi-seed evaluation, and the joint training experiments would all have been impractical at TFT training cost.

The evaluation requires two design choices that turn out to matter. Several wells in the Brandenburg network share identical surface coordinates and are forced into the training set; without this co-location constraint, a random test split can place a test well at the same physical location as a training well, at which point the GP copies a nearby value rather than interpolating, and the measured error does not reflect genuine spatial generalization. The primary evaluation uses a random 90/10 split (104 test wells) repeated over 40 seeds, each drawing a new random 90/10 partition of wells. Running 40 seeds is necessary because seeds differ substantially in how spatially isolated their test wells are, and that isolation is the dominant predictor of per-seed RMSE (Section 5.6).

One of the central findings is about where the bottleneck lies. The two-stage pipeline (GP conditioned on GRU predictions) and a true-observation interpolation baseline (GP conditioned on true observed GWL) differ in per-well RMSE by only 0.003 m ($p = 0.50$, meaning a difference of this size would arise by chance half the time even if the true gap were zero; Section 5.4.2). The GRU’s forecast quality is not the limiting factor; the GP’s ability to generalize across spatial locations is. At $h = 16$, the two-stage pipeline achieves a mean per-well RMSE of 1.082 m, roughly $12\times$ the GRU’s temporal RMSE of 0.088 m at monitored wells. A systematic GP feature ablation shows that using hydrogeological zone alone is the most beneficial configuration. The reason is visible in per-zone error: at matched nearest-training-well distances, recharge zones are roughly $7\times$ harder to interpolate than discharge zones (Section 5.8.1), which means zone membership encodes spatial structure that coordinates alone cannot capture.

The jointly trained model, in which GRU and GP are trained end-to-end with a combined loss, is worse on average (1.259 m vs. 1.082 m mean per-well RMSE at $h = 16$). The aggregate hides a distance gradient, though: the two-stage pipeline is better for test wells close to the training network, while the jointly trained model gains an advantage beyond roughly 4.5 km from the nearest training well. In the Brandenburg monitoring network, most test wells fall within 2 km of a training neighbor, which is why the aggregate favors the two-stage pipeline. For genuinely isolated locations, the jointly trained model is the stronger choice.

The research question this thesis starts with is whether a dense monitoring network, combined with a spatial interpolation step, can support a forecast-anywhere system. For locations within a few kilometers of the training network, the answer from the Brandenburg data is yes, with median RMSE well under 1 m for the closest test wells. For genuinely isolated locations the picture is less encouraging: error increases sharply with distance, and joint training only partially closes that gap. Distance to the nearest training well is the dominant constraint, not the quality of the temporal model. The results show what a dense, decades-long monitoring record can achieve and where the hard limits are. Getting reliable predictions far from any monitoring well is still an open problem, and the directions in Section 7.2 are where that work is most likely to advance. Across most of Brandenburg, monitoring wells are close enough that reliable forecasts are already achievable at the distances reported here. Where monitoring is genuinely sparse, the distance-error relationship quantified in this thesis provides a direct basis for prioritizing where new monitoring infrastructure would reduce forecast uncertainty most.

7.2. Future Work

Neural Processes for end-to-end spatiotemporal modeling. The two-stage spatiotemporal pipeline separates temporal and spatial learning by design. Convolutional Conditional Neural Processes (Gordon et al., 2020) and related architectures can predict at any new location given a set of observed wells as context, without separating temporal and spatial learning into two stages. These represent the most principled alternative to the two-stage approach and should be evaluated on the same Brandenburg test set protocol.

Multi-GRU-seed evaluation. The current evaluation uses a single GRU seed. Including three to five GRU seeds in the full 40-seed spatial evaluation matrix would fully characterize the relative contributions of temporal model variance and spatial split variance to the reported uncertainty.

Multi-seed hyperparameter optimization. Hyperparameter search in this thesis uses single-seed evaluations, which have high variance due to the spatial-split sensitivity documented in Section 4.2.9. Running each trial over multiple spatial splits would reduce that noise and produce a more reliable ranking between candidate architectures. However, because the dominant bottleneck is GP spatial generalization rather than GRU model capacity, better hyperparameter

search is unlikely to substantially improve end-to-end interpolation error; the main benefit would be methodological, not a path to dramatically lower RMSE. The same applies to the GP and to the jointly trained model’s hyperparameter search. For joint training specifically, which is the component where performance remains furthest behind, improved HPO may offer more practical improvement than refining the GRU configuration further.

Improved joint training architecture. The joint training experiment adapts the two-stage architecture for end-to-end training rather than designing a new architecture from scratch. The two-stage pipeline outperforms it at most distances, and an architecture designed from scratch for joint training might narrow that gap.

Generalization to other monitoring networks. All results are derived from a single study area. Replication on other dense groundwater monitoring networks (e.g., the Netherlands Grondwaterregister, the USGS groundwater monitoring network, or the monitoring networks of other German states) would test whether the optimal GP feature set and the interpolation baseline gap magnitude are specific to Brandenburg’s hydrogeology or are transferable.

Integration of physically derived spatial features. The GP relies on a single static feature, hydrogeological zone, selected by the feature ablation in Section 5.3. At genuinely unmonitored locations, hydrogeological zone may not be available at the required spatial resolution and would need to be interpolated from regional geological maps. Quantifying how such interpolated zone assignments affect GP predictions is a necessary step toward deployment at locations with no monitoring record.

Removing per-well normalization. The pipeline z-scores each well before training, which makes the GP responsible for reconstructing the spatial mean at test locations (Section 3.2). Kunz et al. (2025) skip this step and use GWL standard deviation as a static input instead. Whether the GIST pipeline works better without per-well normalization is a straightforward ablation to run.

Spatial block holdout for stricter generalization testing. The split strategies evaluated in this thesis draw test wells from the full monitoring network. A harder test would hold out all wells within a contiguous geographic subregion, with no training wells present in that region. Spatial block cross-validation of this kind is the recommended approach for evaluating spatial models when autocorrelation would otherwise inflate apparent performance (Roberts et al., 2017), and applying it here would assess whether the GRU-GP pipeline generalizes to areas beyond its training region boundaries rather than interpolating between scattered training wells.

Co-located well selection. Right now all co-located wells are forced into training to avoid trivially easy or misleadingly hard test cases (Section 3.4.1). Keeping only the topmost well from each co-located group eligible for the test set would expand the test pool at the cost of a small reduction in training data, without reintroducing those problems.

Operational weather inputs and forecast uncertainty. All experiments in this thesis use HYRAS 5.0 reanalysis weather data, which is available only retrospectively. In an operational deployment, precipitation and temperature forecasts over 16 weeks carry substantial uncertainty. Quantifying how that uncertainty propagates into GWL predictions, and whether it changes the relative ranking of the pipeline configurations, is a necessary step toward actual deployment.

GP uncertainty calibration. The GP’s 95% prediction intervals achieve only approximately 60% empirical coverage (PICP) at spatial test wells, reflecting that the GP underestimates its own uncertainty rather than making poor point predictions. Improving this is important if the pipeline is used for risk assessment or decision support rather than point forecasting alone. Spatially adaptive kernels that account for local differences in monitoring density between training and test regions are a natural starting point.

Empirical Bayesian kriging. Zowam and Milewski (2024) replace a global variogram with a local estimate fitted around each prediction point, reducing sensitivity to the stationarity assumption. Adapting this to the GP stage could improve interpolation in regions where the spatial covariance structure differs from the network-wide average, given that zone-specific error varies by nearly an order of magnitude (Section 5.8.1).

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A. Hyperparameter Configurations

A.1. Best GRU Configuration (Two-Stage Spatiotemporal Pipeline)

Hyperparameter	Value
Hidden dimension	256
Number of layers	4
Dropout	0.2
Learning rate	2×10^{-4}
Batch size	4096
Training epochs	50
Early stopping patience	5
Input window length	52 weeks
Output horizon	16 weeks

A.2. Best GP Configuration (Two-Stage Spatiotemporal Pipeline)

Hyperparameter	Value
Kernel	Matérn-3/2, ARD length-scales
Inference	Exact (Cholesky factorization)
Mean function	Zero
GP features	Hydrogeological zone
GP pretraining	128 randomly sampled points (all dates, grouped per horizon)
Kernel optimization	300 gradient-descent steps on marginal log-likelihood
GP conditioning dates	All test-period dates

A.3. Best Jointly Trained Spatiotemporal Model Configuration

Hyperparameter	Value
GRU hidden / layers	256 / 4
GRU dropout	0.2
Learning rate	2×10^{-4}
Batch size	per-date
Early stopping patience	15
GP loss weight λ	1×10^{-4}
GP conditioning dates	all test-period dates

B. Extended Ablation Results

B.1. GP Feature Ablation Results

All 30 configurations sorted by nRMSE_{pw} . The selected configuration (hydrogeological zone only) is marked with a star.

Table B.1.: GP feature ablation: all 30 configurations sorted by nRMSE_{pw} (median per well, mean over 10 split seeds, 1 GP seed; random 90/10 split; coordinate-only baseline $\text{nRMSE}_{\text{pw}} = 5.077$, $\text{RMSE}_{\text{pw}} = 1.130$ m)

Feature configuration	nRMSE_{pw}	ΔnRMSE	RMSE_{pw} (m)
Hydrogeological zone only *	4.444	−0.633	1.052
Screen top depth only	4.883	−0.194	1.070
Screen bottom depth only	4.889	−0.188	1.075
Both screen depths	4.906	−0.170	1.072
Confinement only	5.017	−0.060	1.148
<i>Spatial coordinates only (baseline)</i>	5.077	—	1.130
Residence time + hydrogeological zone	5.118	+0.041	1.198
GWL class only	5.155	+0.078	1.256
Hydrogeological zone + confinement	5.207	+0.130	1.153
52-week precipitation lag only	5.392	+0.316	1.267
Residence time only	5.437	+0.361	1.261
Residence time + confinement	5.481	+0.404	1.315
ACF class only	5.735	+0.658	1.332
Residence time + both screen depths	5.794	+0.717	1.444
Residence time + screen bottom	5.800	+0.723	1.442
Residence time + screen top	5.800	+0.723	1.453
Residence time + hydro-zone + confinement	5.863	+0.786	1.357
52-week precipitation cross-corr only	6.040	+0.963	1.473
1-week precipitation lag only	6.126	+1.049	1.278
Res. time + hydro-zone + confinement + screen bot- tom	6.193	+1.116	1.454
Res. time + hydro-zone + confinement + both screen depths	6.226	+1.149	1.430
Res. time + hydro-zone + confinement + screen top	6.267	+1.190	1.427

(continued on next page)

Feature configuration	nRMSE _{pw}	Δ nRMSE	RMSE _{pw} (m)
Res. time + hydro-zone + confinement + GWL class (one-hot)	6.443	+1.366	1.586
Res. time + hydro-zone + confinement + GWL class (cont.)	6.598	+1.521	1.595
Res. time + hydro-zone + confinement + 52w precip lag	6.791	+1.714	1.616
Residence time + ACF class	6.800	+1.723	1.624
Res. time + hydro-zone + confinement + 52w precip xcorr	6.901	+1.824	1.726
Res. time + hydro-zone + confinement + ACF class	6.903	+1.826	1.643
Res. time + hydro-zone + confinement + 1w precip lag	6.950	+1.873	1.548
Res. time + hydro-zone + confinement + all precip	7.350	+2.273	1.815

B.2. Spatial Split Comparison

Metric	Dense-area	Cluster-stratified	Sparse-area	Random 90/10*	Random 80/20
Test wells	104	98	104	104	208
Seeds	10	10	10	40	10
avg RMSE _{pw} (m)	0.678 $\pm$ 0.089	1.215 $\pm$ 0.202	5.102 $\pm$ 0.690	1.119 $\pm$ 0.208	1.240 $\pm$ 0.154
h = 16 RMSE _{pw} (m)	0.681 $\pm$ 0.109	1.164 $\pm$ 0.220	5.409 $\pm$ 0.803	1.082 $\pm$ 0.193	1.200 $\pm$ 0.142
avg nRMSE _{pw}	2.900 $\pm$ 0.379	5.123 $\pm$ 0.973	22.269 $\pm$ 3.671	4.580 $\pm$ 0.909	5.084 $\pm$ 0.586
h = 16 nRMSE _{pw}	2.755 $\pm$ 0.329	4.749 $\pm$ 0.964	23.064 $\pm$ 5.054	4.399 $\pm$ 0.988	4.798 $\pm$ 0.541
avg NSE _{pw}	-13.5 $\pm$ 3.2	-42.8 $\pm$ 18.1	-878.9 $\pm$ 262.1	-38.4 $\pm$ 16.9	-45.4 $\pm$ 14.1

* Primary evaluation (40 seeds, hydrogeological zone GP features). All other rows: GRU seed 40, GP seed 1.

Table B.2.: Two-stage pipeline across split strategies, $f=0.90$

B.3. Additional Tables

Split seed	Interp. Baseline RMSE _{pw} (m)	Two-Stage RMSE _{pw} (m)	Gap (m)
ss42	1.127	1.122	−0.005
ss43	1.021	0.998	−0.023
ss44	1.162	1.126	−0.036
ss45	0.974	1.014	+0.039
ss46	0.818	0.827	+0.009
ss47	1.029	1.048	+0.019
ss48	1.045	1.050	+0.005
ss49	1.420	1.380	−0.040
ss50	1.001	1.063	+0.062
ss51	0.842	0.808	−0.034
ss52	1.138	1.159	+0.021
ss53	1.302	1.338	+0.036
ss54	1.212	1.223	+0.011
ss55	1.124	1.118	−0.007
ss56	0.872	0.905	+0.033
ss57	0.913	0.918	+0.006
ss58	1.025	1.046	+0.020
ss59	1.314	1.301	−0.013
ss60	1.375	1.366	−0.008
ss61	1.120	1.123	+0.004
ss62	0.934	0.896	−0.038
ss63	1.385	1.352	−0.032
ss64	1.060	1.068	+0.008
ss65	0.896	0.909	+0.013
ss66	1.224	1.270	+0.046
ss67	0.751	0.751	+0.000
ss68	1.131	1.136	+0.005
ss69	1.060	1.050	−0.011
ss70	1.268	1.281	+0.013
ss71	1.182	1.191	+0.010
ss72	0.975	0.934	−0.041
ss73	1.601	1.617	+0.017
ss74	0.890	0.937	+0.047
ss75	1.228	1.102	−0.125
ss76	1.195	1.213	+0.017
ss77	0.827	0.847	+0.020
ss78	1.020	0.999	−0.021
ss79	1.593	1.597	+0.005
ss80	1.478	1.518	+0.040
ss81	1.096	1.140	+0.045
Mean	1.116	1.119	+0.003
Std	0.208	0.208	0.033

Table B.3.: Per-seed interp. baseline vs. two-stage RMSE_{pw}, rand 90/10

Split seed	avg RMSE _{pw} (m)	h = 16 RMSE _{pw} (m)	avg NSE _{pw}	h = 16 NSE _{pw}
ss42	1.122	1.128	−31.5	−23.0
ss43	0.998	0.974	−21.9	−19.7
ss44	1.126	1.057	−44.1	−41.1
ss45	1.014	1.117	−40.7	−23.5
ss46	0.827	0.912	−33.7	−23.4
ss47	1.048	1.065	−46.4	−30.6
ss48	1.050	0.918	−31.5	−16.8
ss49	1.380	1.251	−38.6	−32.5
ss50	1.063	1.000	−31.2	−32.7
ss51	0.808	0.846	−29.7	−29.2
ss52	1.159	1.129	−45.7	−38.6
ss53	1.338	1.340	−48.6	−47.4
ss54	1.223	1.071	−25.5	−18.6
ss55	1.118	1.157	−34.4	−42.2
ss56	0.905	0.887	−15.2	−13.5
ss57	0.918	0.858	−22.1	−18.1
ss58	1.046	1.053	−33.5	−33.1
ss59	1.301	1.089	−30.0	−31.5
ss60	1.366	1.240	−42.8	−30.9
ss61	1.123	1.223	−32.8	−34.2
ss62	0.896	0.828	−19.2	−15.8
ss63	1.352	1.313	−72.2	−54.3
ss64	1.068	1.101	−40.0	−27.5
ss65	0.909	0.902	−21.0	−18.5
ss66	1.270	1.186	−43.1	−35.8
ss67	0.751	0.742	−18.3	−13.9
ss68	1.136	1.153	−31.4	−24.5
ss69	1.050	0.986	−27.2	−20.5
ss70	1.281	1.204	−49.9	−48.0
ss71	1.191	1.202	−50.3	−42.7
ss72	0.934	0.912	−25.6	−26.8
ss73	1.617	1.496	−72.7	−73.4
ss74	0.937	0.790	−26.6	−21.2
ss75	1.102	1.049	−52.4	−43.3
ss76	1.213	1.128	−32.3	−24.1
ss77	0.847	0.846	−32.6	−30.0
ss78	0.999	0.919	−21.1	−18.8
ss79	1.597	1.511	−75.6	−62.7
ss80	1.518	1.536	−92.6	−92.3
ss81	1.140	1.159	−50.2	−41.2
Mean	1.119	1.082	−38.4	−32.9
Std	0.208	0.193	16.9	16.4

Table B.4.: Two-stage pipeline per-seed metrics, rand 90/10

Split seed	avg RMSE _{pw} (m)	h = 16 RMSE _{pw} (m)	avg NSE _{pw}	h = 16 NSE _{pw}
ss42	1.127	1.101	−32.7	−19.5
ss43	1.021	1.008	−22.7	−19.5
ss44	1.162	1.049	−45.2	−35.8
ss45	0.974	1.135	−42.9	−27.3
ss46	0.818	0.883	−30.4	−25.1
ss47	1.029	1.008	−46.4	−30.9
ss48	1.045	0.892	−31.5	−16.4
ss49	1.420	1.248	−39.9	−32.5
ss50	1.001	0.967	−35.1	−30.9
ss51	0.842	0.875	−30.4	−29.1
ss52	1.138	1.136	−44.6	−36.0
ss53	1.302	1.317	−48.3	−47.9
ss54	1.212	1.099	−26.1	−18.6
ss55	1.124	1.148	−36.6	−42.1
ss56	0.872	0.860	−15.4	−12.8
ss57	0.913	0.852	−22.3	−18.7
ss58	1.025	1.041	−31.9	−31.2
ss59	1.314	1.092	−27.9	−28.1
ss60	1.375	1.244	−41.6	−34.5
ss61	1.120	1.209	−32.2	−31.8
ss62	0.934	0.853	−19.0	−15.3
ss63	1.385	1.358	−74.9	−61.5
ss64	1.060	1.085	−39.6	−29.2
ss65	0.896	0.865	−21.0	−18.3
ss66	1.224	1.210	−42.7	−38.0
ss67	0.751	0.734	−19.7	−12.7
ss68	1.131	1.170	−32.2	−24.0
ss69	1.060	1.008	−27.7	−22.3
ss70	1.268	1.184	−51.3	−50.8
ss71	1.182	1.145	−48.6	−41.1
ss72	0.975	0.968	−25.2	−25.2
ss73	1.601	1.514	−69.8	−72.3
ss74	0.890	0.825	−26.4	−22.7
ss75	1.228	1.128	−53.3	−49.9
ss76	1.195	1.125	−31.8	−24.8
ss77	0.827	0.841	−34.1	−30.1
ss78	1.020	0.893	−20.8	−20.0
ss79	1.593	1.476	−76.0	−64.6
ss80	1.478	1.511	−95.6	−92.4
ss81	1.096	1.084	−50.6	−41.6
Mean	1.116	1.079	−38.6	−33.1
Std	0.208	0.191	17.1	16.9

Table B.5.: True-observation interpolation baseline per-seed metrics, rand 90/10

Split seed	avg RMSE _{pw} (m)	h = 16 RMSE _{pw} (m)	avg NSE _{pw}	h = 16 NSE _{pw}
ss42	1.088	1.053	−35.0	−32.9
ss43	1.406	1.379	−41.4	−39.2
ss44	1.134	1.133	−48.7	−38.7
ss45	1.149	1.132	−36.6	−30.8
ss46	1.199	1.340	−46.6	−52.5
ss47	1.070	1.062	−41.8	−31.8
ss48	0.976	0.903	−27.3	−25.9
ss49	1.378	1.046	−65.4	−42.4
ss50	0.946	1.038	−24.4	−21.6
ss51	1.163	1.119	−42.2	−43.8
ss52	1.195	1.163	−53.8	−43.3
ss53	1.563	1.748	−80.8	−81.0
ss54	1.522	1.586	−56.4	−53.3
ss55	1.140	0.995	−30.5	−26.7
ss56	1.065	0.996	−16.9	−13.9
ss57	1.007	1.093	−26.9	−21.1
ss58	0.976	0.881	−21.3	−17.9
ss59	1.224	1.276	−42.0	−43.2
ss60	1.495	1.393	−49.9	−45.4
ss61	1.613	1.647	−47.0	−51.0
ss62	1.090	1.134	−45.5	−39.6
ss63	1.361	1.125	−61.6	−43.0
ss64	1.604	1.532	−82.6	−73.9
ss65	1.427	1.364	−69.1	−59.8
ss66	1.549	1.535	−44.6	−40.5
ss67	1.072	1.122	−28.8	−30.3
ss68	1.157	1.013	−35.2	−26.5
ss69	1.290	1.324	−51.7	−43.7
ss70	1.587	1.585	−61.3	−52.8
ss71	1.158	1.038	−42.9	−39.0
ss72	1.150	1.193	−50.0	−48.7
ss73	1.702	1.716	−100.3	−86.3
ss74	1.008	1.069	−33.1	−21.4
ss75	1.377	1.390	−86.0	−77.6
ss76	1.582	1.177	−60.6	−37.2
ss77	1.202	1.212	−47.0	−47.1
ss78	0.947	0.986	−30.1	−25.8
ss79	1.528	1.484	−67.6	−57.5
ss80	2.319	2.347	−218.8	−205.2
ss81	1.028	1.011	−32.7	−30.5
Mean	1.286	1.259	−52.1	−46.1
Std	0.278	0.290	32.8	30.8

Table B.6.: Jointly trained model per-seed metrics, rand 90/10

Table B.7.: Joint model λ sweep (h256/l4, 1 GP seed)

λ	RMSE _{pw} (m)	nRMSE _{pw}
0.9	1.403	7.88
0.7	1.352	8.58
0.5	1.255	6.27
0.3	1.048	5.89
0.1	1.091	5.62
0.01	1.115	5.42
0.001	1.025	5.50
0.0001	1.016	5.50
0.00001	1.348	7.22

Table B.8.: Joint model architecture sweep ($\lambda = 0.034$, 1 GP seed), h256/l4 selected

Arch	Hidden	Layers	RMSE _{pw} (m)	nRMSE _{pw}
h256/l4	256	4	1.419	7.08
h128/l5	128	5	1.426	7.17

Metric	Two-Stage	Jointly Trained
<i>Per-well spatial metrics (mean $\pm$ std, 40 seeds)</i>		
h = 16 RMSE _{pw} (m)	1.082 $\pm$ 0.193	1.259 $\pm$ 0.290
avg RMSE _{pw} (m)	1.119 $\pm$ 0.208	1.286 $\pm$ 0.278
avg nRMSE _{pw}	4.580 $\pm$ 0.909	—
h = 16 NSE _{pw}	−32.9 $\pm$ 16.4	−46.1 $\pm$ 30.8
avg NSE _{pw}	−38.4 $\pm$ 16.9	−52.1 $\pm$ 32.8
NSE _{pw} P10	−3731 $\pm$ 1858	−3299 $\pm$ 1871
NSE _{pw} P25	−505 $\pm$ 300	−532 $\pm$ 279
NSE _{pw} P75	−3.83 $\pm$ 1.93	−6.49 $\pm$ 4.12
NSE _{pw} P90	−0.221 $\pm$ 0.353	−0.580 $\pm$ 0.804
RMSE _{pw} P10 (m)	0.243 $\pm$ 0.036	0.268 $\pm$ 0.051
RMSE _{pw} P25 (m)	0.440 $\pm$ 0.071	0.548 $\pm$ 0.131
RMSE _{pw} P50 (m)	1.119 $\pm$ 0.208	1.286 $\pm$ 0.278
RMSE _{pw} P75 (m)	3.497 $\pm$ 0.740	3.624 $\pm$ 0.826
RMSE _{pw} P90 (m)	9.015 $\pm$ 2.382	8.871 $\pm$ 1.974
<i>Uncertainty calibration</i>		
PICP-95	0.602 $\pm$ 0.043	0.629 $\pm$ 0.055
<i>Global (pooled) metrics</i>		
Global RMSE (m)	7.169 $\pm$ 1.502	6.243 $\pm$ 1.258
Global NSE	0.917 $\pm$ 0.029	0.937 $\pm$ 0.022
Global nRMSE	0.219 $\pm$ 0.056	0.192 $\pm$ 0.051
Global MAE (m)	3.437 $\pm$ 0.635	3.291 $\pm$ 0.557
Global KGE	0.909 $\pm$ 0.045	0.927 $\pm$ 0.036
Abs. error P95 (m)	15.43 $\pm$ 3.15	13.60 $\pm$ 3.06
Abs. error P99 (m)	30.82 $\pm$ 9.99	25.64 $\pm$ 8.17

Table B.9.: Two-stage vs. joint: full metrics, rand 90/10

Split seed	Nearest-training-well dist. (km)	avg RMSE _{pw} (m)	h = 16 RMSE _{pw} (m)
ss62	1.718	0.896	0.828
ss58	1.720	1.051	1.053
ss51	1.783	0.810	0.846
ss67	1.806	0.750	0.742
ss81	1.812	1.145	1.159
ss64	1.966	1.072	1.101
ss56	1.974	0.911	0.887
ss77	1.990	0.852	0.846
ss44	1.991	1.125	1.057
ss57	1.997	0.918	0.858
ss43	2.008	1.002	0.974
ss54	2.010	1.225	1.071
ss65	2.020	0.910	0.902
ss72	2.046	0.936	0.912
ss42	2.064	1.122	1.128
ss45	2.080	1.017	1.117
ss76	2.105	1.227	1.128
ss55	2.115	1.123	1.157
ss59	2.139	1.303	1.089
ss78	2.172	1.001	0.919
ss49	2.174	1.379	1.251
ss50	2.183	1.063	1.000
ss47	2.184	1.051	1.065
ss63	2.188	1.353	1.313
ss69	2.203	1.048	0.986
ss74	2.208	0.944	0.790
ss68	2.216	1.136	1.153
ss80	2.231	1.519	1.536
ss66	2.240	1.275	1.186
ss46	2.246	0.826	0.912
ss48	2.257	1.051	0.918
ss71	2.292	1.191	1.202
ss61	2.317	1.121	1.223
ss75	2.323	1.105	1.049
ss60	2.331	1.365	1.240
ss70	2.339	1.285	1.204
ss73	2.352	1.627	1.496
ss79	2.386	1.595	1.511
ss52	2.422	1.159	1.129
ss53	2.537	1.337	1.340
Spearman ρ (dist. vs. avg RMSE _{pw})		0.603 ($p < 0.001$)	
Spearman ρ (dist. vs. h = 16 RMSE _{pw})		0.601 ($p < 0.001$)	

ρ computed from full-precision values; displayed distances are rounded to 3 d.p. All RMSE_{pw} values are medians over test wells.

Table B.10.: Per-seed two-stage pipeline RMSE_{pw} vs. median nearest-training-well distance (random 90/10, 40 seeds, sorted by nearest-training-well distance)

Decile	NTN range (km)	NTN median (km)	n pairs	Two-stage median RMSE _{pw} (m)	Joint median RMSE _{pw} (m)	Two-stage wins (%)
D1 (closest)	0.01–0.61	0.394	422	0.675	1.098	66.8
D2	0.62–0.99	0.823	420	0.566	0.909	64.5
D3	1.00–1.36	1.220	411	0.699	0.771	60.3
D4	1.36–1.72	1.570	420	0.608	0.950	63.8
D5	1.73–2.25	2.043	414	0.725	0.860	51.4
D6	2.25–2.72	2.456	409	0.899	0.836	51.8
D7	2.72–3.36	3.036	420	1.481	1.534	51.7
D8	3.37–4.24	3.776	413	1.853	1.953	40.2
D9	4.24–5.45	4.677	418	2.406	2.550	45.7
D10 (farthest)	5.45–21.4	6.809	413	5.369	4.261	31.7

Table B.11.: Two-stage vs. jointly trained by NTN distance decile

Table B.12.: Top 5 static features by permutation importance (run 0346, h256/l4, 5 repeats per feature). Delta is the mean increase in nRMSE_{pw} relative to the baseline (0.0467). All 35 features have positive delta; only the five largest are shown.

Rank	Feature	$\Delta \text{nRMSE}_{\text{pw}}$	% change
1	Pardé seasonality coefficient	+0.0221	+47.3%
2	Water courses fraction	+0.0057	+12.3%
3	Topographic Wetness Index	+0.0049	+10.5%
4	Surface elevation	+0.0048	+10.2%
5	Pastures fraction	+0.0040	+8.6%
—	All remaining 30 features	+0.001–+0.003	+2–+7%

Table B.13.: Dynamic covariate permutation importance (run 0346, h256/l4, 5 repeats per feature). Delta is the mean increase in nRMSE_{pw} relative to the baseline (0.0467).

Rank	Feature	$\Delta \text{nRMSE}_{\text{pw}}$	% change
1	Temperature (5 km)	+0.0428	+91.5%
2	Precipitation (5 km)	+0.0395	+84.5%
3	Calendar cosine encoding	+0.0187	+40.0%
4	Relative humidity (5 km)	+0.0099	+21.2%
5	Calendar sine encoding	+0.0098	+20.9%

C. Static Feature Definitions and Model Assignments

Tables C.1 to C.3 list all 40 static features available in the Brandenburg groundwater dataset, grouped by their role in the modelling pipeline. **GRU** features are passed as `x_static` to the temporal model. **GP** features are used as spatial covariates in the Gaussian Process kernel. Features marked — are available in the dataset but are not primary model inputs in the main experiments.

Table C.1.: GRU static inputs (35 features). All are standardized before training. The three hydrogeological zone indicators are additionally used as GP spatial inputs (Table C.2).

Column name	Group	Description
corine_1000_fractions_ corineclccode_111	Land use	Continuous urban fabric frac- tion
corine_1000_fractions_ corineclccode_112	Land use	Discontinuous urban fabric fraction
corine_1000_fractions_ corineclccode_121	Land use	Industrial/commercial units fraction
corine_1000_fractions_ corineclccode_131	Land use	Mineral extraction sites frac- tion
corine_1000_fractions_ corineclccode_141	Land use	Green urban areas fraction
corine_1000_fractions_ corineclccode_211	Land use	Non-irrigated arable land frac- tion
corine_1000_fractions_ corineclccode_231	Land use	Pastures fraction
corine_1000_fractions_ corineclccode_242	Land use	Complex cultivation patterns fraction
corine_1000_fractions_ corineclccode_311	Land use	Broad-leaved forest fraction
corine_1000_fractions_ corineclccode_312	Land use	Coniferous forest fraction
corine_1000_fractions_ corineclccode_313	Land use	Mixed forest fraction
corine_1000_fractions_ corineclccode_321	Land use	Natural grasslands fraction

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Column name	Group	Description
corine_1000_fractions_ corineclccode_322	Land use	Moors and heathland fraction
corine_1000_fractions_ corineclccode_411	Land use	Inland marshes fraction
corine_1000_fractions_ corineclccode_511	Land use	Water courses fraction
corine_1000_fractions_ corineclccode_512	Land use	Water bodies fraction
TWI_dgm50_r1000m	Terrain	Topographic Wetness Index (1 km buffer)
gok	Terrain	Surface elevation (m a.s.l.)
eumohp_1000_mean_dsd1	EU-MOHP	Drainage system density, Strahler-1 watercourses
eumohp_1000_mean_lp1	EU-MOHP	Linear permeability, Strahler- 1
eumohp_1000_mean_sd1	EU-MOHP	Stream density, Strahler-1
amatulli_1000_mean_ entgeom10kment	Geomorpho90m	Geomorphological entropy (10 km scale)
amatulli_1000_mean_ shannongeom10kmsha	Geomorpho90m	Shannon entropy of terrain (10 km scale)
amatulli_1000_mean_ unigeom10kmuni	Geomorpho90m	Terrain uniformity (10 km scale)
huek250_1000_fractions_kf_9	HÜK250	Hydraulic conductivity class 9 fraction
huek250_1000_fractions_kf_ 10	HÜK250	Hydraulic conductivity class 10 fraction
huek250_1000_fractions_kf_ 11	HÜK250	Hydraulic conductivity class 11 fraction
gwn1000_1000_mean_recharge	GW recharge	Mean annual GW recharge, source 1 (mm/yr)
GW_recharge_r1000m	GW recharge	Mean annual GW recharge, source 2 (mm/yr)
parde_seasonality	Seasonality	Pardé seasonality coefficient (per-well time series)
gw_gespannt_bin	Confinement	Aquifer confinement binary (1 = confined)

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Column name	Group	Description
siwa_verweilzeit_j	Residence time	GW residence time in years (ordinal: 0,1,3,10,30,50)
hydroraum_Speisungsgebiete	Hydro zone	Recharge zone indicator (binary)
hydroraum_Transitgebiete	Hydro zone	Transit zone indicator (binary)
hydroraum_Entlastungsgebiete	Hydro zone	Discharge zone indicator (binary)

Table C.2.: GP spatial inputs (3 features). One-hot encoding of hydrogeological zone (**hydroraum**); also included in the GRU feature set above.

Column name	Group	Description
hydroraum_Speisungsgebiete	Hydro zone	Recharge zone indicator (binary)
hydroraum_Transitgebiete	Hydro zone	Transit zone indicator (binary)
hydroraum_Entlastungsgebiete	Hydro zone	Discharge zone indicator (binary)

Table C.3.: Features available in the dataset but not used as primary model inputs.

Column name	Group	Description
gwlk	Aquifer complex	Glacial aquifer complex (3 classes: <i>Grundwasserleiterkomplexe</i>)
fok	Screen depth	Well screen top depth, <i>Filteroberkante</i> (~19% missing)
fuk	Screen depth	Well screen bottom depth, <i>Filterunterkante</i> (~19% missing)
acf_class	ACF	Autocorrelation lag class (<6, <9, <12, <20, <53, >53 weeks); requires historical GWL to compute
gw_gespannt	Confinement	Raw confinement string; replaced by gw_gespannt_bin